

Perfect 2048: Kernel-Verified Optima and Extremal Play

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Abstract

We study extremal *possible play* in 2048: player moves and legal spawn outcomes together specify a trajectory, without asserting that those outcomes can be forced in random play. For every ordinary two-tile opening of the standard 4×4 game, we determine the exact minimum number of rounds to reach 131072: 32,781 from two 4s and 32,782 otherwise. Every opening attains its time optimum jointly with the minimum post-spawn endpoint mass 131102. These results, including coverage of all 480 labeled openings, are checked by Lean 4’s kernel. The constructive theorem has no axiom dependencies; the combined optimum theorem uses only `propext` and `Quot.sound`.

Written bounds and independently replayed witnesses further establish the maximum score 1,966,216 among fastest target-reaching games, and a 65,548-round optimum when exactly one 4-birth is permitted. We certify 320 distinct minimum-length full-chain arrangements in 40 symmetry orbits, including every boundary location of the largest tile and diagonally opposite placements of the smallest and largest tiles. Unavoidable total-mass configurations yield policy-independent probability bounds, a fixed spawn-value word for any ceiling-score game, and a scalar relaxation of optimal expected score. We also give a uniform one-dimensional construction and a moving-window embedding lemma.

The broader score and game-length bounds are checked on enumerated small boards: the 2×5 possible-play computation has 1,814,373,350 positions up to symmetry and attains 36,828 points and 2,034 rounds; the cube and the published 3×3 count supply further comparisons. Attainment of the 4×4 score ceiling 3,932,100 remains conjectural. The Lean scope is the fixed-board time–mass optima and supporting deterministic lemmas, not all stochastic, score, or arbitrary-board arguments. Plaintext witnesses, two independent replayers, proof sources, and reproducible audits accompany the manuscript.

1 Introduction

2048 [4] is played on a 4×4 board. A move slides all tiles in one of four directions; equal tiles that collide merge into their sum, each tile merging at most once per move; then one new tile appears in a uniformly random empty cell, a 2 with probability 0.9 and a 4 with probability 0.1. The score is the sum of all merged values. The game ends when no move changes the board.

Alok Menghrajani’s fork [27] adds an undo button that re-randomises the spawn on replay. That small change turns the game into a different object: a player who is willing to press undo often enough can realise *any* spawn sequence, and so can play any game that is possible in principle. This paper is about those games. Five questions organise it:

1. What is the fewest moves in which the largest tile can be built?
2. What is the highest score any game can reach?
3. What is the longest any game can last?
4. What must the final position of such a game look like, and in how many ways can it be arranged?
5. What does the undo button cost, in presses, to play such a game?

The first two questions have folklore answers. For the move count, the observation that every move adds a spawned 2 or 4 to the board gives a lower bound immediately; the cascade that finishes the tile adds a few more moves, and it was not obvious that the sum is exact. For the score, several numbers circulate: 3,932,164 (pure mass accounting), 3,932,156 (two “structurally forced” 4-spawns), and 3,932,100 (one forced 4-spawn per tile of the final position). Our own earlier README argued for a ceiling of 3,932,156 and claimed geometry prevents paying it. That claim was half right: geometry does prevent it, but the right ceiling is 3,932,100, and we prove it. Even the largest tile has a history: the tile-count argument that caps it at 131072 is old, but whether the spatial rules let a game actually reach it was doubted in 2014 [34] and still called open in 2017 [20]; Das and Paul [7] state it for boards of any shape, by an argument with a gap we discuss in Section 10. The line shipped with this paper is an explicit certificate for the standard game, of the minimum possible length, published as a text file that a short independent program checks (Section 6.4). The same witness and a universal lower bound are now kernel-checked in Lean 4, along with the exact 32,782-move optimum for two initial 2s (Section 6.5). The numerical score and longest-game figures already appear in Ripà’s 2014 analysis [32, 33]. We contribute explicit proof arguments and certificates rather than claiming those numbers as new.

Contributions.

- Theorem 20: a Lean-kernel-checked legal 32,781-move trajectory to 131072, a lower bound over all ordinary openings and legal plays, and the exact 32,782-move optimum when both initial tiles are 2s. The formal semantics, threshold-mass argument, certificate generation, axiom audit, and reproducible build are described in Section 6.5.
- Theorem 4: 2^n needs at least $(2^n - 8)/4 + (n - 2)$ moves on any board, with equality for $n = 11$ (519) and $n = 17$ (32,781) on 4×4 and for every relevant tile on each enumerated board of at most ten cells. Lemma 5: every fastest top tile passes through one and the same full board and finishes with a forced cascade.
- Theorem 9: on any c -cell board, score $\leq \Phi_c - 4c$. The argument identifies a merge tree’s cell requirement with its Strahler number (the Ershov register count) and shows that the c final tiles must pay their tight moments in a forced order.
- Theorem 14: no game has more than $2^{c+1} - 4 - c$ moves, and the longest games are the maximum-score games up to their last spawn. Proposition 15: every game ending on the full chain ends the same way, which pins the 4 to the boundary beside the 8.
- Theorem 22: the rules commute with the eight symmetries of the square, the corner snakes form a single orbit, and one computed line is therefore eight perfect games—one per corner and direction.
- Exact values (Section 5.5): every bound above is attained on 2×2 , 2×3 , 2×4 , 3×3 , 2×5 and the $2 \times 2 \times 2$ cube; the enumerator agrees with the published 3×3 state count.
- Section 6: the algorithm that computed the 4×4 lines and the certificate that makes its output a theorem (Theorem 19): 32,781 and 65,533 moves, optimal, and 129,333 moves for 3,925,224 points with 1,735 spawned 4s where the ceiling allows 16; we locate where the 4s go and conjecture the ceiling is attainable.

- Section 8: the exact expected number of undo presses for each shipped line under honest odds, and the opening problem.
- Section 9: a strong solution of the small boards (expected score, tile probabilities) and an expectimax player for the full board measured against the classical heuristics and an adversarial tile generator.

2 Related work

Complexity and combinatorics. Langerman and Uno [17] showed that deciding whether a given position can be played to a given tile is NP-hard on $m \times n$ boards for Threes, 1024 and 2048; Abdelkader, Acharya and Dasler [1] showed the variant without new tiles is still hard; Mehta [26] gave a PSPACE-completeness result for a related decision problem. Eppstein [9] analysed the maximum score in a relaxed model of 2048-like games in which any set of cells whose values sum to a tile value may be merged in one step, proving a min-max theorem that equates the maximum score with the smallest value for which greedy change-making needs a given number of coins; the binary-counter strategy every strong 2048 player follows is, in that light, greedy change-making in base 2. The relaxed model has only the small spawn value. The real game’s 4-spawns are precisely the resource that buys an extra cell, and Theorem 9 prices that resource: four points each, and exactly c of them on the way to the top position.

Numerical prior art and scope. Ripà [32, 33] already gives the numerical score formula, the 131052-round figure, and the necessary rare spawn-value word probability in 2014. A contemporaneous account [2] observes that the final 2 or 4 does not affect the score. We retain this credit when proving bounds and correcting the equality case; we do not treat successful computation or formalization as evidence of historical priority. No uniform arbitrary-rectangle construction or 4×4 score-ceiling witness is asserted here.

Learning agents. Szubert and Jaśkowski [38] introduced temporal-difference learning of n -tuple networks for 2048; Wu et al. [39] added multi-stage learning; Jaśkowski [14] reached the strongest reported agents; Matsuzaki and co-authors studied the systematic selection of n -tuples and neural-network players [28, 23, 24, 25]; Guei, Wei and Wu [13, 12] used 2048-like games for teaching reinforcement learning and proposed optimistic TD learning. Slizkov [37] proved rigorous bounds on honest play: a strategy reaching 2048 with probability at least 0.99969, and 256 with certainty.

Exact solving. Lees-Miller [21, 19, 22] analysed the game with Markov chains and Markov decision processes, computed the minimum of 519 moves to the 2048 tile, and enumerated the 2×2 and 3×3 games. Yamashita, Kaneko and Nakayashiki [41] strongly solved 3×3 (48,713,519 positions) and Kaneko and Yamashita [15] strongly solved 4×3 (1,152,817,492,752 positions, 739,648,886,170 afterstates, optimal expected score about 50,724.26 from a two-2s start), the key technique being to partition the state space by the sum of tile values (the “age” of a state). Our enumerator uses the same grading and, on 3×3 , reproduces their position count exactly (Section 5.5).

Higher dimensions. Das and Paul [7] analysed 2048 on $n_1 \times n_2$ boards and on d -dimensional boxes $n_1 \times \dots \times n_d$: their Theorem 1 states that the largest tile 2^{c+1} , c the number of cells, can be reached on every such board, and they give strategies for the tile-placing adversary and, under

conditions, for the player. Rodgers and Levine [34] had called 131072 the maximum attainable tile “in theory” while doubting the video evidence for it, and Kohler, Migler and Khosmood [16] repeat the 2^{n^2+1} figure. Our arguments use nothing of a board beyond its cell count and the fact that a slide acts line by line, so Theorems 4, 9 and 14 and Proposition 15 hold verbatim in every dimension, and Section 10 solves Das and Paul’s $2 \times 2 \times 2$ cube exactly: every bound is attained there too.

Recent preprints. Saligram, Bhathal and Manihani [35] compare four deep reinforcement learning agents (DQN, PPO, QR-DQN and a distributional “Horizon-DQN”); the preprint reports the last reaching a 4096 tile with a best score of 41,828, and has since been withdrawn by its authors pending regenerated results. Bai, Kim Cohen, Koss and Lichtenbaum [3] evolve 2048 players with large language models, either by meta-prompting a “thinker”/“executor” pair, which did not improve, or by refining the value function of a limited Monte Carlo tree search, which gained about 473 points per cycle. An independent strong solution of the 3×3 board [30] reports an optimal expected score of 5,468.5 and tile probabilities of 99.571%, 73.677% and 1.134% for 256, 512 and 1024, which agree with Table 2 to every digit shown.

What is new here. We distinguish numerical observations, abstract bounds, explicit legal trajectories, and formal verification. Lees-Miller’s 519 is the length of a path in a Markov chain built for the 4×4 board by always taking the 4; Theorem 4 proves the formula behind it for every board and every tile, and Lemma 5 shows that every fastest top tile passes through the same board. Eppstein’s min–max theorem is for a relaxed game with only 2-spawns and arbitrary set merges; in the real game the spawned 4s buy cells, and Theorem 9 prices them, which is why the answer is $\Phi_c - 4c$ and not a change-making number. The longest game (Theorem 14), the forced ending of every full-chain game (Proposition 15), the eight-spirals statement (Theorem 22), the exact undo expectations (Proposition 24) and the exact small-board values of Table 1 are, as far as we can tell, new; the reachability of the top tile, claimed in general by Das and Paul, gets an independently checkable certificate on 4×4 together with the exact fewest moves and a Lean-kernel-checked proof (Section 6.5), and is joined by score and length ceilings that hold in any dimension, all attained on the first three-dimensional board solved; the maximum-score figures 3,932,164 and 3,932,156 that circulate are shown to be unattainable; and the 3×3 count of Yamashita et al. is independently reproduced. The snake strategy in one corner is folklore among players and the target of every heuristic program; that its eight images are all realised by one transformed move list is, to our knowledge, not stated anywhere, though it will surprise no implementer.

3 Preliminaries

A board is a finite set of c cells with a slide structure (rows and columns in the rectangular case; most arguments below use only the cell count). Tiles carry values 2^k , $k \geq 1$; we call k the *rank*. A *play* is a sequence of moves, each consisting of a slide (which performs its merges) followed by a *spawn* of a 2 or a 4 into an empty cell. The game starts with two spawns. Let n_2 and n_4 be the numbers of spawned 2s and 4s so far, counting the two starting tiles. A position is *full* when every cell holds a tile; on a full board a line (row or column) in which no merge is available cannot move.

Definition 1 (possible play). In *possible play* the spawn values and cells are chosen by the player. Every finite spawn sequence has positive probability in the honest game, so a position is reachable

in possible play if and only if it is reachable with positive probability in the honest game; and a player with a re-randomising undo button can realise any possible play.

For a position B write $\text{mass}(B) = \sum_{t \in B} 2^{k_t}$ and

$$\Phi(B) = \sum_{t \in B} (k_t - 1) 2^{k_t}.$$

Lemma 2 (mass ledger). *Slides conserve mass and every spawn adds its value: $\text{mass}(B) = 2n_2 + 4n_4$ at all times. After m moves the mass is at most $8 + 4m$, with equality if and only if every spawn so far, the two starting tiles included, was a 4. Since every move spawns exactly once, the number of moves of any play reaching B is*

$$m = n_2 + n_4 - 2 = \frac{1}{2}\text{mass}(B) - n_4 - 2.$$

Lemma 3 (score identity). *At every moment of every play, $\text{score} = \Phi(B) - 4n_4$.*

Proof. By induction over events. Initially the score is 0; a starting 2 contributes 0 to Φ , a starting 4 contributes 4 and raises $4n_4$ by 4. A merge of two 2^k tiles into 2^{k+1} scores 2^{k+1} and changes Φ by $k2^{k+1} - 2(k-1)2^k = 2^{k+1}$. A spawned 2 changes nothing on either side; a spawned 4 adds 4 to Φ and 4 to $4n_4$. $\square$

So every spawned 4 costs exactly four points relative to what the final position “looks like it is worth”, and the maximum score of a game is $\max_B (\Phi(B) - 4n_4^{\min}(B))$ over reachable positions B , where $n_4^{\min}(B)$ is the fewest spawned 4s on any play reaching B . By the same token the *longest* play reaching B is the one with the fewest 4s: two plays to the same position with the same number of spawned 4s have the same length.

Merge trees. Every tile on the board is the root of a binary *merge tree*: its leaves are spawned tiles (value 2 or 4), each internal node is a tile created by merging its two children. A tile of rank k has all leaves at depth $k-1$ (2-leaves) or $k-2$ (4-leaves). Two facts about the game drive everything below:

- *Persistence.* A tile, once spawned or created, stays on the board until it merges into its parent. Nothing else removes it.
- *One doubling per move.* A tile’s value at most doubles in a move, because each tile merges at most once per move.

Call the tiles present on the board that belong to the merge tree of a future tile T the *pieces* of T . By persistence, from the first spawn of a leaf of T until T is created, at least one piece of T is on the board; afterwards T itself is.

4 The fewest moves

Theorem 4. *On any board and for any play, the tile 2^n ($n \geq 3$) does not exist before move*

$$M(n) = \frac{2^n - 8}{4} + (n - 2).$$

Equality is attainable only if the game starts from two 4s and every spawn outside the last $n-2$ moves is a 4.

Proof. Let T be the first move after which a tile 2^n exists. A tile spawned at move $T - j$ has value at most $4 = 2^2$ then, and by one doubling per move its value at move T is at most 2^{2+j} ; for it to be part of the 2^n we need $2 + j \geq n$, i.e. $j \geq n - 2$. Hence the $n - 2$ tiles spawned at moves $T - (n - 3), \dots, T$ are not part of the 2^n : they are still on the board (persistence), as themselves or merged among each other, so the position after move T has mass at least 2^n plus the sum of those $n - 2$ spawn values.

Write z for the number of spawned 2s among the T move-spawns and z' for the number among the last $n - 2$ of them, so $z' \leq z$. By Lemma 2 the mass after move T is $M_0 + 4T - 2z$ with $M_0 \leq 8$ the starting mass, and the last $n - 2$ spawns sum to $4(n - 2) - 2z'$. Therefore

$$\begin{aligned} M_0 + 4T - 2z &\geq 2^n + 4(n - 2) - 2z', & \text{so} \\ 4T &\geq 2^n - 8 + 4(n - 2) + 2(z - z') + (8 - M_0) \geq 2^n - 8 + 4(n - 2), \end{aligned}$$

which is the claim. Equality forces $M_0 = 8$ and $z = z'$. $\square$

Attainability is a construction. For $n = 11$, Lees-Miller's Markov chain [21] exhibits the $519 = 510 + 9$ move win on 4×4 . For $n = 17$ we generated, verified and ship a 32,781-move line (Section 6); Table 1 shows that the bound is attained for each relevant tile on each enumerated board up to ten cells—the 2048 tile, for instance, needs exactly 519 moves on the ten-cell 2×5 board too, where it is the largest tile there is. For the largest tile a board can hold, the fastest plays are far more constrained than the theorem lets on: they all pass through the same tile multiset. The fixed- 4×4 reachability and lower-bound claims are formalised in Section 6.5; that section also handles the two-2s opening.

Lemma 5 (the primed board). *Let a play on a board with $c \geq 2$ cells create the tile 2^{c+1} at move $M(c+1)$. Then after move $M(c+1) - (c-1) = (2^{c+1} - 8)/4$ the board is full and holds exactly the tiles $2^c, 2^{c-1}, \dots, 8, 4, 4$, every one of them a piece of the coming 2^{c+1} . In each of the remaining $c - 1$ moves exactly one merge of these pieces takes place, creating $8, 16, \dots, 2^{c+1}$ in this order, and none of the $c - 1$ tiles spawned during those moves ever joins the 2^{c+1} .*

Proof. Put $n = c + 1$, $T = M(n)$ and $t_0 = T - (n - 2)$. By the equality clause of Theorem 4 the play starts from two 4s and every spawn up to move t_0 is a 4, so after move t_0 the mass is $8 + 4t_0 = 2^n$. As in the proof of the theorem, a leaf of the merge tree of the 2^n spawned at move $T - j$ needs $j \geq n - 2$, so every leaf is spawned at or before move t_0 . After move t_0 each leaf is contained in exactly one piece on the board (persistence), so the pieces present have total mass 2^n —the mass of the whole board. Every tile is therefore a piece, and every tile, being built from 4s, is a power of two of value at least 4.

The tile spawned at move t_0 is a 4 that must double in each of the moves $t_0 + 1, \dots, T$. The remaining tiles are powers of two, each at least 4, summing to $2^n - 4 = 4(2^{n-2} - 1)$; in units of 4 they are powers of two summing to $2^{n-2} - 1$, whose binary expansion has $n - 2$ ones, and a sum of powers of two has at least as many terms as the binary expansion of its value has ones. So there are at least $n - 2 = c - 1$ of them, at least c tiles in all on c cells: exactly c , the board is full, and equality in the binary bound means the $c - 1$ remaining tiles are exactly $2^c, \dots, 8, 4$.

At move $t_0 + 1$ the spawned 4 merges with the only other 4 into an 8; at move $t_0 + 2$ that 8 must double again, and the only other 8 is the primed one; inductively the k -th of the last $c - 1$ moves merges the new 2^{k+1} with the primed 2^{k+1} . No tile spawned after t_0 can take part: after $k - 1$ such spawns their total mass is at most $4(k - 1) < 2^{k+1}$. All other primed tiles are distinct singletons, so no other merge among pieces is possible. $\square$

For 4×4 the lemma says that every 32,781-move win shows, after move 32,766, the full board 65536, 32768, $\dots$, 8, 4, 4 (Figure 1, right), and then plays a cascade of fifteen forced merges $4 + 4$, $8 + 8$, $\dots$, $65536 + 65536$. The lemma leaves the *arrangement* free; the fifteen junk spawns of the cascade are the only freedom in the values. This is exactly what the “proven-minimal 15-move fold” of Section 6 realises. For tiles below the top one the pieces may be split further—at move 510 of a 519-move win on 4×4 the board holds pieces of the coming 2048 of total mass 2048, but they need not be 1024, $\dots$, 8, 4, 4: eleven tiles 1024, $\dots$, 16, 4, 4, 4, 4 can fold in the same nine moves by merging two pairs of 4s at once. On 2×5 , where 2048 is the top tile, the lemma applies and every 519-move win passes through 1024, $\dots$, 8, 4, 4.

Corollary 6. *On a board with c cells the full chain $2^{c+1}, 2^c, \dots, 4$ (one distinct power of two per cell, a dead position) needs at least $2^c - 3$ moves, and that many only if every spawn is a 4. On 4×4 this is 65,533, and it is attained.*

Proof. Its mass is $2^{c+2} - 4$; by Lemma 2 at least $(2^{c+2} - 12)/4$ moves are needed and every spawn must be a 4. The shipped line (Section 6) does it in exactly 65,533 moves; along it the 131072 first forms at move 32,784, three later than $M(17)$: all-4 feeding needs a few consolidations that a 2-spawn during the cascade would avoid, and the ledger absorbs them. $\square$

5 The highest score, the longest game, and the last move

5.1 A merge tree needs its Strahler number of cells

The *Strahler number* of a binary tree is $S(\text{leaf}) = 1$ and, for a node with subtrees A, B , $S = \max(SA, SB)$ if they differ and $SA + 1$ if they are equal. It is the number of registers needed to evaluate an expression tree without spilling (Ershov [10], Sethi–Ullman [36], Flajolet–Raoult–Vuillemin [11]) and the black pebbling number of the tree [29]. Building a tile is evaluating its merge tree with cells as registers, and the persistence of pieces makes the lower bound short to prove directly.

Lemma 7. *Let T be a tile of rank $k \geq 2$ with merge tree $\mathcal{T}$. Then $S(\mathcal{T}) = k$ if every leaf is a 2, and $S(\mathcal{T}) = k - 1$ if at least one leaf is a 4.*

Proof. Induction on k . For $k = 2$: a spawned 4 is a leaf ($S = 1$), and $2 + 2$ has $S = 2$. For $k > 2$ the children are trees of rank $k - 1$ with Strahler numbers in $\{k - 2, k - 1\}$, equal to $k - 1$ exactly when all their leaves are 2s. Two all-2 children give $S = k$; otherwise the maximum is $k - 1$ and, if both are $k - 2$, $S = k - 1$ as well. $\square$

Lemma 8 (tight moment). *During the construction of a tile with merge tree $\mathcal{T}$ there is a moment at which at least $S(\mathcal{T})$ pieces of $\mathcal{T}$ are on the board simultaneously.*

Proof. Induction on $\mathcal{T}$; a leaf is its own piece. Let the root have subtrees A and B , roots created at times $t_A < t_B$ (if they are created in the same move, either order works), merged at time $t^* > t_B$. Let τ_A, τ_B be tight moments of A, B given by induction, so $\tau_A \leq t_A$ and $\tau_B \leq t_B$. If $SA > SB$, or $SB > SA$, the larger side’s tight moment already shows $\max(SA, SB)$ pieces. Suppose $SA = SB = s$; we need $s + 1$.

- If $\tau_B > t_A$: at τ_B the s pieces of B are on the board and so is the finished root of A (created at $t_A < \tau_B < t_B \leq t^*$, and persistent): $s + 1$ pieces.
- If $\tau_B \leq t_A$ and $\tau_A > t_B$: symmetric, s pieces of A plus the root of B .

- If $\tau_B \leq t_A$ and $\tau_A \leq t_B$: from τ_B until t_B at least one piece of B is on the board (persistence), and $\tau_A \leq t_A < t_B$; if $\tau_A \geq \tau_B$ then at τ_A there are s pieces of A and at least one of B . If instead $\tau_A < \tau_B \leq t_A$, then at τ_B there are s pieces of B and at least one piece of A (present from τ_A until $t_A \geq \tau_B$).

In every case some moment shows $s + 1$ pieces of $\mathcal{T}$. $\square$

5.2 The ceiling

Let $\Phi_c = \sum_{k=2}^{c+1} (k-1)2^k = (c-1)2^{c+2} + 4$, the value of Φ on the full chain $2^{c+1}, 2^c, \dots, 2^2$.

Theorem 9. *On any board with $c \geq 2$ cells, every play satisfies $\text{score} \leq \Phi_c - 4c$. Equality after a complete round requires the full chain with exactly c spawned 4s, or the same chain with its 4 replaced by a 2 and exactly $c-1$ spawned 4s. Each of the $c-1$ larger tiles has exactly one spawned 4 in its merge tree; a final 4 is itself spawned.*

Proof. Fix a play and its final position with tiles $T_1, \dots, T_m$, $m \leq c$, of ranks $k_1, \dots, k_m$. Every spawned tile of the whole play is a leaf of exactly one T_i , so $n_4 = \sum_i h'_i$ where h'_i counts the 4-leaves of T_i ; put $h_i = \min(h'_i, 1)$. By Lemma 7 the Strahler number of T_i 's tree is $S_i = k_i - h_i$ (a tile of rank 1 is a spawned 2, $S_i = 1$).

Let τ_i be a tight moment of T_i (Lemma 8) and relabel so that $\tau_1 \leq \tau_2 \leq \dots \leq \tau_m$. At τ_p the board holds S_p pieces of T_p and, for each $q < p$, at least one piece of T_q or T_q itself (persistence, since $\tau_q \leq \tau_p$). Hence $S_p + (p-1) \leq c$, i.e.

$$k_p \leq c + 1 - p + h_p \quad (p = 1, \dots, m). \quad (1)$$

By Lemma 3,

$$\text{score} = \sum_p (k_p - 1)2^{k_p} - 4n_4 \leq \sum_p [(k_p - 1)2^{k_p} - 4h_p].$$

For fixed p the bracket, subject to (1), is largest with $h_p = 1$ and $k_p = c + 2 - p$, worth $(c + 1 - p)2^{c+2-p} - 4$; the alternative $h_p = 0$, $k_p = c + 1 - p$ is worth $(c - p)2^{c+1-p}$, which is smaller for $p < c$ and equal ($= 0$) for $p = c$. All the terms are nonnegative, so filling the board ($m = c$) is best, and

$$\text{score} \leq \sum_{p=1}^c [(c + 1 - p)2^{c+2-p} - 4] = \sum_{k=2}^{c+1} (k-1)2^k - 4c = \Phi_c - 4c.$$

Equality forces $m \geq c - 1$, and for $p < c$ it forces $k_p = c + 2 - p$ and $h'_p = h_p = 1$. If $m = c - 1$, every final tile has value at least 8, which is impossible just after the mandatory 2/4 spawn. Thus $m = c$. The final zero-valued bracket permits both $(k_c, h_c, h'_c) = (2, 1, 1)$ and $(1, 0, 0)$, giving the two stated cases. $\square$

Corollary 10. *On 4×4 , $\text{score} \leq \Phi_{16} - 64 = 3,932,164 - 64 = 3,932,100$. The numbers 3,932,164 and 3,932,156 are not attainable.*

Remark 11. The theorem uses only the number of cells, persistence, and one merge per tile per move. It holds for any adjacency structure, any spawn rule that offers the values 2 and 4, and in particular in possible play. The constraint that actually binds is that a tile of rank k built from 2s alone needs k cells at once, and the c tiles of the top position must take their tight moments in decreasing order of size on a board that the larger ones are already sitting on.

Inequality (1) is worth more than the score bound. It bounds the ranks of the final tiles by their tight-moment order, and so it says what the top tile and the top position are and what they cost.

Corollary 12 (the top tile). *On a board with c cells no tile exceeds 2^{c+1} . Every tile 2^{c+1} has a spawned 4 among its leaves, and at some moment during its construction every cell of the board holds one of its pieces.*

Proof. A tile of rank k has Strahler number k or $k - 1$ (Lemma 7) and needs that many cells at its tight moment (Lemma 8), so $k - 1 \leq c$; for $k = c + 1$ the Strahler number must be c , which forces a 4-leaf, and the tight moment fills the board. $\square$

That 131072 is the largest tile of 4×4 2048 is folklore, and Das and Paul [7] state the bound, and claim attainability, for boards of any shape and dimension; this is a two-line proof of the bound that says a little more: the 131072 is never built from 2s alone, and there is a moment when it owns the whole board.

Corollary 13 (the top position). *On a board with c cells:*

- (i) *the full chain is the unique tile multiset of maximum mass, $2^{c+2} - 4$;*
- (ii) *every play reaching the full chain has at least c spawned 4s, one in the merge tree of each of its tiles; in particular its 4 is itself a spawned 4, and the chain cannot be built from 2s alone;*
- (iii) *a play reaching the full chain has between $2^c - 3$ and $2^{c+1} - 4 - c$ moves.*

Proof. By (1), $k_p \leq c + 2 - p$, so $\text{mass} = \sum_p 2^{k_p} \leq \sum_{p=1}^c 2^{c+2-p} = 2^{c+2} - 4$ with equality only for the ranks $c + 1, c, \dots, 2$: (i). For the chain $k_p = c + 2 - p$ forces $h_p = 1$ in (1): (ii). By Lemma 2 the number of moves is $\frac{1}{2}(2^{c+2} - 4) - n_4 - 2$ with $c \leq n_4 \leq m + 2$: (iii). $\square$

5.3 The longest game

Theorem 14 (the longest game). *On a board with c cells no play has more than $2^{c+1} - 4 - c$ moves. Equality requires the final position to be the full chain reached with exactly c spawned 4s, or the full chain with its 4 replaced by a 2 reached with exactly $c - 1$ spawned 4s; in either case the score is $\Phi_c - 4c$. On 4×4 the bound is 131,052.*

Proof. By Lemma 2 the length is $\frac{1}{2}\text{mass} - n_4 - 2$. With the notation of the proof of Theorem 9, $n_4 \geq \sum_p h_p$ and (1) holds, so

$$\text{moves} \leq \sum_{p=1}^m (2^{k_p-1} - h_p) - 2.$$

For $p < c$ the bracket is at most $2^{c+1-p} - 1$ (take $h_p = 1$; the alternative $h_p = 0$ gives 2^{c-p} , which is smaller); for $p = c$ both choices give 1; and every bracket is positive, so $m = c$ is best. Summing, $\text{moves} \leq \sum_{p=1}^c (2^{c+1-p} - 1) - 2 = 2^{c+1} - 4 - c$. Equality forces $k_p = c + 2 - p$ and $h_p = 1$ for $p < c$, $(k_c, h_c) \in \{(2, 1), (1, 0)\}$, and $n_4 = \sum_p h_p$. The score follows from Lemma 3: the final 2 contributes nothing to Φ and saves one 4. $\square$

So the longest game and the highest-scoring game are the same game up to its last spawn, and Conjecture 16 below is equivalent to the 4×4 longest game lasting 131,052 moves. The shipped maximum-score line, at 129,333 moves, is the longest 4×4 game we know of.

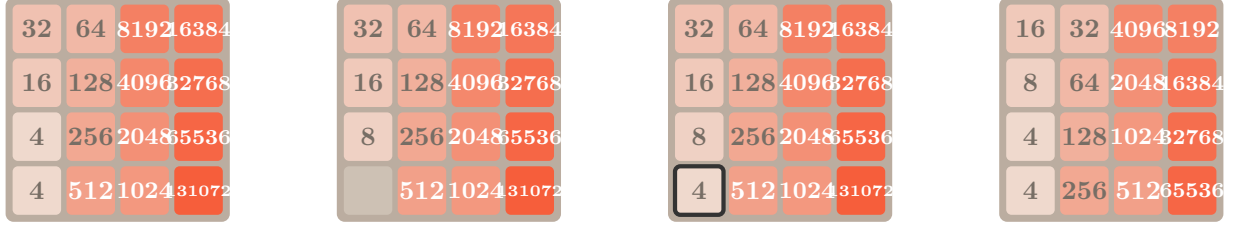


Figure 1: Left three boards: the last move of every full-chain game, here in the application's default orientation (bottom-right corner, chain running up). The board is full with two 4s; the slide (up) merges them into the 8 and frees the far end of the column; the final 4 spawns there. Right: the primed board of Lemma 5, the position after move 32,766 of every 32,781-move win, in the same arrangement.

5.4 The last move

Proposition 15 (the last move). *Let a play on a rectangular board with $c \geq 3$ cells end on the full chain after move L . Then:*

- (i) *the spawn of move L is a 4, and it is the 4 of the final position;*
- (ii) *the slide of move L is played from a full board holding $2^{c+1}, \dots, 16$ and two 4s; it merges exactly one pair, $4 + 4 \rightarrow 8$, and moves nothing outside that line;*
- (iii) *the spawn of move $L - 1$ is also a 4;*
- (iv) *the 4 of the final position occupies the end cell of the line of the last slide, opposite to the direction of the slide—a boundary cell—and the 8 lies in the same line.*

Proof. Let F be the final multiset and s_t the value spawned at move t . The tile spawned at move L is on the final board as itself, so $s_L \in F \cap \{2, 4\} = \{4\}$: (i). Before that spawn the board held $F \setminus \{4\}$, $c - 1$ tiles. If the slide of move L merged j pairs, the position P after move $L - 1$ held $c - 1 + j$ tiles, so $j \leq 1$; and P contains the tile $s_{L-1} \in \{2, 4\}$. If $j = 0$ then $P = F \setminus \{4\}$, which has no tile of value at most 4: impossible. So $j = 1$, P is full, and $P = F \setminus \{4, 2^k\} \cup \{2^{k-1}, 2^{k-1}\}$ for the merged pair; the only tile of P that can be at most 4 is 2^{k-1} , whence $k = 3$, $P = \{2^{c+1}, \dots, 16, 4, 4\}$ and $s_{L-1} = 4$: (ii), (iii). On a full board a line without a merge cannot move; the merging line is full, so its two 4s are adjacent, and after the merge its three tiles pack toward the wall the slide moves to, leaving the far end cell empty—the only empty cell on the board, where the final 4 spawns: (iv). $\square$

Proposition 15 is verified exhaustively on every board of Table 1 with at most ten cells—all 16, 112, 912, 4,224 and 11,136 moves that produce the full chain on 2×2 , 2×3 , 2×4 , 3×3 and 2×5 , and all 9,216 on the $2 \times 2 \times 2$ cube, have this form—and on the shipped lines, whose last three moves are $4 + 4 \rightarrow 8$, $8 + 8 \rightarrow 16$, $4 + 4 \rightarrow 8$ with four spawned 4s in a row (Figure 1). On 4×4 it says that in *every* reachable arrangement of the full chain the 4 is one of the twelve boundary cells and shares a row or column with the 8. The snake arrangements of Section 7 put the 4 at a corner with the 8 beside it.

5.5 Exact values on small boards

The mass of a position grows by exactly the spawned value each move, so the state graph is a directed acyclic graph graded by mass, and can be enumerated one layer at a time with only the two open layers in memory. Kaneko and Yamashita [15] call this grading the age of a state. We

board	c	positions	(raw)	max score	$\Phi_c - 4c$	$\Phi_c - 8$	chain in	longest	$2^{c+1} - 4 - c$
2×2	4	110	662	180	180	188	13	24	24
2×3	6	21,752	85,844	1,260	1,260	1,276	61	118	118
2×4	8	4,980,767	19,920,382	7,140	7,140	7,164	253	500	500
3×3	9	48,713,519	388,921,077	16,352	16,352	16,380	509	1,011	1,011
$2 \times 2 \times 2$	8	702,581	32,187,234	7,140	7,140	7,164	253	500	500
2×5	10	1,814,373,350	7,257,192,913	36,828	36,828	36,860	1,021	2,034	2,034
4×4	16	?	? $\geq$ 3,925,224	3,932,100	3,932,156	65,533	$\geq$ 129,333	131,052	

Table 1: Exact extremes of possible play. “positions” counts reachable positions up to the board’s symmetries (8 for squares, 4 for rectangles, 48 for the cube), “raw” without reduction; “chain in” is the fewest moves to the full chain ($2^c - 3$, Corollary 6); “longest” the longest game (Theorem 14). On every enumerated board up to ten cells, the three-dimensional one included, the maximum score equals the ceiling of Theorem 9, the full chain is reached with exactly c spawned 4s, and the longest game equals the bound. The 3×3 count agrees with Yamashita et al. [41, 15]. The 4×4 row holds the bounds of this paper and the shipped lines.

board	$\mathbb{E}[\text{score}]$, standard start	given a two-2s start	max probability of the largest tiles
2×2	66.96	67.70	16: 96.25%, 32: 8.29%
2×3	480.26	480.99	64: 93.93%, 128: 5.70%
2×4	2,641.87	2,642.60	256: 86.98%, 512: 2.46%
3×3	5,468.49	5,469.21	512: 73.68%, 1024: 1.13%
$2 \times 2 \times 2$	2,953.42	2,954.15	256: 94.04%, 512: 6.32%

Table 2: Strong solutions of the honest game (uniform empty cell, 90% twos). The expected score is under the score-maximising policy; each tile probability is under the policy that maximises that probability. The two-2s column uses Kaneko and Yamashita’s convention (their 4×3 figure is 50,724.26). The 3×3 row agrees with the independent solution of [30]. The cube has six directions, and with the same eight cells is a better board than the 2×4 strip.

implemented it (`test/solve.js`) with positions reduced by the board’s symmetry group and, per position, the fewest spawned 4s and the fewest moves on any play reaching it; the maximum score is then $\max_B(\Phi(B) - 4n_4^{\min}(B))$ by Lemma 3 and the longest game is $\max_B(\frac{1}{2}\text{mass}(B) - n_4^{\min}(B) - 2)$ by Lemma 2. The same layered pass, run backwards, strongly solves the honest game: the optimal expected score and, for each tile, the maximum probability of ever building it.

Table 1 summarises. On every board tested the score ceiling is attained, always on the full chain, always with exactly one spawned 4 per tile, and the longest game is exactly $2^{c+1} - 4 - c$ moves and ends on the full chain or on the chain with a 2 in place of the 4, as Theorem 14 predicts; the fewest moves to every tile 2^k with $k \geq 3$ equals $M(k)$ of Theorem 4, and the fewest to the full chain equals $2^c - 3$. The ten-cell 2×5 board, with $1.8 \cdot 10^9$ positions up to symmetry, took three hours in a forward-only pass that keeps only the open layers. Our implementation of the stochastic backward pass retained all layers and exceeded available memory; that stochastic solution was not completed. This is an implementation limitation, not a necessity to retain all layers in RAM. The 3×3 enumeration gives 48,713,519 positions, the number Yamashita, Kaneko and Nakayashiki report, which also fixes the convention of their count: positions up to the eight symmetries of the square. (Lees-Miller’s “about 25 million” for 3×3 counts differently.) The honest-game side of the same computation is in Table 2.

5.6 The 4×4 board

The 4×4 state space is out of reach of enumeration (Lees-Miller’s month-long run [19] enumerated $1.3 \cdot 10^{12}$ states without finishing; the full game is far larger). What we have is a constructive lower bound. The generator of Section 6 walks a binary counter along the snake path into a corner, feeding 2s, and grants a spawned 4 only when a bounded look-ahead shows every pure-2 continuation dying. It produced a line of 129,333 moves ending on the full chain with $n_4 = 1,735$, hence a score of

$$\Phi_{16} - 4 \cdot 1,735 = 3,932,164 - 6,940 = 3,925,224,$$

99.825% of the ceiling and 6,876 points short of it. The identities $\text{moves} = 131,068 - n_4$ (all 129,333 moves spawn 2 except the 4s; $\text{mass } 262,140 = 4 + 2 \cdot \text{moves} + 2n_4$) and $\text{score} = 3,932,164 - 4n_4$ are verified by replay through the game engine in all four corners and both spiral orientations.

Where do the 1,735 go? The first act—building 131072 on the empty board—spends exactly one, at its tight moment. The whole gap is in the second act, the chain 65536, $\dots$, 4 on the fifteen cells beside the 131072: 295 spawned 4s while building the 65536 and 1,439 below it, most of them with three to five empty cells still on the board. So the generator is not being squeezed by the pebbling bound; it is being squeezed by geometry within a cascade. When the counter overflows, the resulting chain of merges takes one move per level, a tile spawns during each of those moves, and the spawns land in the cells the cascade has just freed, alongside the tail of the chain. The bounded look-ahead cannot always see how to digest that junk with 2s alone and buys its way out with a 4. Theorem 9 says the price of the second act is at least fifteen 4s; the small boards, which contain the same corner turns and the same overflow cascades in miniature, all pay exactly the theorem’s price.

Conjecture 16. *The maximum score in 4×4 2048 is 3,932,100, attained by a play with exactly sixteen spawned 4s; equivalently (Theorem 14), the longest 4×4 game has 131,052 moves.*

Two routes to settle it suggest themselves. The direct route is a better search for the second act: the target is a schedule that, at each level k of the chain, builds the first copy of 2^{k-1} from 2s on the $k - 1$ free cells (which Lemma 7 permits) and the second copy with a single 4 at its own tight moment, and that handles the overflow junk explicitly. The indirect route is Kaneko and Yamashita’s enumeration: their layered pass over 4×3 can carry a two-byte “fewest 4s” field at negligible cost and would settle the 4×3 maximum, which Theorem 9 bounds by $\Phi_{12} - 48 = 180,180$, and the 4×3 longest game, bounded by $2^{13} - 16 = 8,176$; we predict equality in both.

6 The perfect algorithm

The lines the application plays were not found by searching the game at run time. They were computed once, offline, by a program that knows in advance how long its line must be, checked by an independent replay, and shipped as data (`js/perfect_line.js`, two hex characters per move: direction, spawn cell, spawn value); the browser replays them. This section describes the construction and, more to the point, what is *proved* about its output. The heuristics need no proof: they only decided whether a line was found. The theorems of the previous sections decide whether the line found is optimal.

6.1 The length is fixed before the search starts

Lemma 2 turns the two move-minimal targets into fixed-length problems. Under all-4 spawns the mass after m moves is $8 + 4m$, so the primed board of Lemma 5 (mass 131,072) is reached after exactly 32,766 moves on *every* route that reaches it at all, and the full chain (mass 262,140) after exactly 65,533. The tile count is pinned too: two tiles at the start, one more per spawn, one fewer per merge, sixteen on the primed board, so the build performs exactly $2 + 32,766 - 16 = 32,752$ merges in its 32,766 moves—a carry on all but fourteen of them. The generator therefore never optimises length. It has to find a route on which every spawn is a 4, and the ledger does the rest; the program asserts $8 + 4 \cdot \text{steps} = \text{mass}$ at every checkpoint and refuses any line that overruns the ledger. For the maximum-score line the roles reverse: spawns are 2s wherever possible, the only free quantity is n_4 , and by Lemmas 2 and 3 $\text{moves} = 131,068 - n_4$ and $\text{score} = 3,932,164 - 4n_4$: the line is judged by one number.

6.2 The snake and the counter

Definition 17 (corner snake). Fix a corner κ of the 4×4 board and one of the two sides through it. The *corner snake* $\sigma(\kappa, \text{side}) = (S_0, \dots, S_{15})$ is the boustrophedon path that starts at κ , runs along the chosen side, and turns back at every wall: it visits the four lines parallel to that side in order, alternating direction. Figure 3 shows all eight. The *chain along* σ is the full chain arranged with 2^{17-i} on S_i (Figure 2).

The construction keeps the occupied prefix of the snake a binary counter: values non-increasing along $S_0, S_1, \dots$, with at most one adjacent equal pair (a pending carry), and everything past the prefix small feed material. A spawn on the first free snake cell increments the counter, a slide toward the head of the line being filled performs a carry, and when a line fills, the carry turns the corner into the next line. The evaluation is

$$H(B) = \sum_{i < \ell} v_i 2^{-i} - 10^3 \sum_{i \geq \ell} v_i,$$

where v_i is the value on S_i and ℓ is the length of the counter prefix (which must start with the largest tile on S_0): a tile in the counter is worth its value discounted by its depth, every tile outside it is a liability of a thousand times its value.

At each step the generator considers three of the four directions—toward the head of the line being filled, toward the corner’s own line, and away from the head—and, for the 2-fed line only and only when at most two cells are empty, the fourth, which the L-shaped assembly of the last line needs. Spawn cells are the first two empty snake cells and the cells one line deeper than the line being filled, where a tile can wait to drop in. The greedy step takes the best candidate if it raises H ; otherwise a rescue search (iterative deepening to depth 10, ten best candidates per node, a fuel of 60,000 nodes) looks for a line that ends above the current H . Every candidate with at most two empty cells is checked five plies ahead for forced death. When no candidate and no rescue exists the position is a *stall*: its family (the tiles of value at least 8 and the number of empty cells) is banned, the walk retreats 8, 16, $\dots$, 2048 moves along its own line—replaying the line from the start restores the board—and continues around the ban. Mass grows every move, so the walk cannot cycle; a cap on the number of stalls bounds the work.

The fold. From the primed board the program searches the cascade of Lemma 5 by iterative deepening over total length: each move either advances the cascade by exactly the next doubling

(a junk merge may ride along) or consolidates junk without touching the cascade, and the values spawned during the fold, 2 or 4, are chosen so that junk never lines up into an accidental merge that would displace the cascade. The primed board fills the board completely, so every junk spawn lands on the cascade’s path; that the fold can nevertheless be done in the fifteen moves the lemma forces, with no consolidation moves at all, is a fact about the snake arrangement that the search establishes. For the tile line the fold is appended after the counter walk; in the full and score lines it happens mid-line, with one added rule—once the 131072 exists nothing may ever sit head-ward of it on the snake, since no smaller tile can merge its way past.

The 2-fed line. With 2s the counter needs one more cell per level (Lemma 7), and the second act, on fifteen cells beside the 131072, is one cell short of that at every turn. The generator opens a *relief valve* only under demonstrated necessity: after twelve stalls without progress (four near the end) the spawn menu admits 4s, every decision still exhausts its pure-2 options first, and the valve closes again after two hundred moves of progress. The two full-board squeezes—entering the fold and the last move—are handled by Proposition 15’s logic in advance: there the 4 is granted outright.

6.3 What is proved about the output

Proposition 18 (certificate). *Let a list of steps $(d_t, x_t, v_t)_{t=1}^L$ (direction, spawn cell, spawn value) be replayed from a permitted two-tile opening, with every $v_t \in \{2, 4\}$. If every slide d_t changes the board and every cell x_t is empty when its spawn is placed, then the list is a legal play. Conditional on that opening and the prescribed directions, its spawn sequence has probability $\prod_t q_{v_t}/e_t > 0$, where e_t is the number of empty cells after slide t and $q_2 = 0.9$, $q_4 = 0.1$; its length, score and final position are then determined by Lemmas 2 and 3.*

Theorem 19 (the shipped lines). *Three lines pass the replay of Proposition 18, in each of the four corners and both orientations of Theorem 22:*

- (a) 32,781 moves from two 4s to a 131072, optimal by Theorem 4; its position after move 32,766 is the primed board of Lemma 5 arranged along the snake;
- (b) 65,533 moves from two 4s to the chain along the snake, optimal by Corollary 6, scoring 3,670,024;
- (c) 129,333 moves from two 2s to the chain along the snake with $n_4 = 1,735$, scoring 3,925,224—within 6,876 points of Corollary 10 and within 1,719 moves of Theorem 14.

Proof. The generator replays the lines in `test/gen_perfect.js` (all four corners, at generation time). The independent replays are `verify/verify2048.py` and `test/replay_engine.js`; `PERFECT=1 node test/run.js` additionally checks the headless runner in either orientation; `test/paper_facts.js` checks the primed board and the last moves. The optimality claims are the theorems cited. $\square$

6.4 The certificate, in a form anyone can check

Proposition 18 is only as good as the replay, so the lines are also published in a form that owes nothing to this repository’s code: three text files in `witness/` list the two starting tiles and then one line per move—the direction slid and the row, column and value of the tile that appeared—and `verify/verify2048.py`, 160 lines of Python with no dependencies, replays them under the rules of the original game. The verifier implements the slide twice, once cell by cell as in Cirulli’s

`game_manager.js` and once line by line, and stops at the first move where the two disagree, where a slide changes nothing, where a new tile is not a 2 or a 4, or where it lands on an occupied cell. On the three witnesses it reports, in under two seconds:

witness	steps verified	score	2s	4s	final position
<code>131072.txt</code>	32,781	1,966,092	8	32,775	131072 in the corner, 13 junk tiles
<code>full-chain.txt</code>	65,533	3,670,024	0	65,535	the full chain, dead
<code>max-score.txt</code>	129,333	3,925,224	127,600	1,735	the full chain, dead

The counts of spawned 2s and 4s include the two starting tiles, and each report ends with “Illegal moves: 0, Illegal spawns: 0, Certificate: VALID”. A second, deliberately different check (`test/replay_engine.js`) feeds the same text files to the original game engine—Cirulli’s `game_manager.js`, `grid.js` and `tile.js`, unmodified, with the random tile replaced by the prescribed one—and confirms the same counts. Any implementation of the rules will do the same job in a few dozen lines; the format is documented in the files themselves. We are not aware of an earlier explicit, independently checkable certificate that the standard 4×4 game reaches 131072, let alone one of the minimum length.

6.5 Kernel-checked reachability and optimality in Lean

The independent replayers in Section 6.4 check a concrete trajectory. We additionally formalise its legality and its minimum length in Lean 4 [8]. The development in `lean-kernel/` uses Lean 4.19.0 and its standard library, without Mathlib. The lower bound is proved directly for the same slide-and-spawn semantics used to check the witness; no connection to an abstract merge-tree model is assumed.

Formal rules and quantifiers. A Board has four rows of four natural-number values, and each coordinate type has exactly four constructors. A row slide removes zeros, merges equal adjacent inputs from the moving edge, and pads with zeros. The evaluator is proved to satisfy a separate declarative `PairMerge` relation: a newly produced tile is an output, not an input to another merge in that slide. The row-length lemma ensures that padding discards no tile. Reversal and transposition implement all four whole-board directions. Separate lemmas prove exact mass conservation under a slide and exact mass increase by the spawned value under a round.

The function `step` accepts a direction and a spawn only if the slide changes the board, the spawn cell is empty after the slide, and the spawn value is 2 or 4. One tile is then inserted. The inductive relation `Plays a T b` means exactly T such rounds from a to b . The predicate `IsInitial` allows any two distinct cells and either permitted value in each. There is no undo, restart, deletion, selective row move, or arbitrary board edit in this relation. As elsewhere in this paper, continuation after obtaining the 2048 tile is allowed, and the count includes the mandatory spawn after the final slide.

Theorem 20 (Lean-certified exact move counts). *For standard 4×4 2048 under possible play:*

- (a) *Some ordinary two-tile opening admits a legal trajectory to 131072 of length 32,781, and every trajectory to 131072 from any ordinary opening has length at least 32,781.*
- (b) *When both opening tiles are required to be 2s, some such opening admits a trajectory of length 32,782, and every trajectory to 131072 from any such opening has length at least 32,782.*

These are minima over permitted openings and legal spawn sequences, not a claim that every initial placement attains the same minimum or that a player can force those spawns.

The named Lean theorems are:

```
Game2048.standard2048_exact_131072
Game2048.standard2048_two_twos_exact
```

Their statements conjoin an existential legal trajectory with a universal lower bound over all legal trajectories, rather than asserting only that a replay program returned `true`.

The lower bound without ancestry bookkeeping. For a board B and a nonnegative threshold k , put

$$H_k(B) = \sum_{x: B(x) > k} B(x).$$

For each slide S , the formal development proves

$$H_{2k}(S(B)) \leq H_k(B). \quad (2)$$

An unmerged tile counted on the left was already counted on the right. A merged tile $v + v$ exceeds $2k$ precisely when each input exceeds k . No input can participate in another merge during that slide. For $k \geq 2$ a new 2 or 4 contributes nothing above $2k$, so (2) also holds for a complete round.

Define $\text{cutoff}(j) = 2^{j+1}$. Iteration over the last j rounds of a trajectory of length $T \geq j$ gives

$$H_{\text{cutoff}(j)}(B_T) \leq H_2(B_{T-j}) \leq \text{mass}(B_{T-j}) \leq M_0 + 4(T - j).$$

If B_T contains 131072, take $j = 15$, whose cutoff is 65536. With $M_0 \leq 8$ this gives $T \geq 32,781$; with two initial 2s, $M_0 \leq 4$ gives $T \geq 32,782$. The Lean proof handles $T < 15$ as well: iteration through all T rounds bounds $H_{\text{cutoff}(T)}(B_T)$ by $H_2(B_0) \leq 8$, contradicting the target tile. The general theorem `Plays.target_deadline` expresses the resulting target-versus-time inequality. The development also proves the power-of-two lower bound on the fixed 4×4 board for ranks at least four and the 519-round lower bound for 2048. It does not assert attainment for arbitrary target sizes or board shapes.

The constructive half. The input is the unchanged plaintext `witness/131072.txt`. An untrusted Python generator proposes the intermediate boards and emits 32,781 individual transition proofs using `decide +kernel`. Each proof checks the actual `step` function, including nontrivial movement and spawn legality. The resulting plays are composed into 258 blocks in 17 generated modules, then into a single trajectory. The kernel checks both the ordinary opening and the occurrence of 131072 in the terminal board. Thus correctness of the Python generator, the AI, and the JavaScript engine is not an assumption of the Lean theorem.

For the two-2s case, put the two 2s at $(3, 2)$ and $(3, 3)$, with zero-based coordinates. One right slide merges them at $(3, 3)$, followed by a 4-spawn at $(3, 2)$. The resulting board is exactly the two-4s opening of the existing certificate. Lean verifies this extra round and composes it with the original play. The derived plaintext witness is `witness/131072-two-twos.txt`. The first construction, together with the direct lower bound, proves part (a) of Theorem 20; the one-round extension and the stronger opening-mass bound prove part (b).

Audit and reproducibility. The original complete kernel audit passed at proof commit 86bd902; the supplementary results are recorded at 03ca3d1 [6]. A subsequent complete CI rebuild at that supplemented commit passed as run 34094095435, checking both exact-optimum theorems. The files `AUDIT.md` and `COROLLARIES.md` in `lean-kernel/` record the provenance. The original witness has SHA-256


```
4f49c437f0946f17b6afb9c4c97c62ef
1a44ee7595ad75888d3806c3291afe9e
```

(the two displayed lines are concatenated). From the repository root,

```
bash lean-kernel/check.sh
```

rebuilds the proof from source, regenerates the concrete transition proofs, checks the supplementary results, and audits the theorem dependencies. The completed core audit reports:

```
Game2048.Cert131072.certified_play: no axioms
Game2048.reachable_131072: no axioms
Game2048.standard2048_exact_131072:
  [propext, Quot.sound]
```

The two-2s exact theorem likewise uses only `propext` and `Quot.sound`. These are standard foundational axioms of Lean, not hypotheses about 2048. The build rejects unfinished proofs and nonstandard axioms, and it does not use native evaluation as a proof oracle. Kernel reduction, the axiom audit, and review of the formal statement serve different purposes [18]. Regression examples test non-chaining merges and invalid spawns; a negative test requires the kernel to reject a fabricated legal move on an empty board. The source hash identifies the witness but is not a substitute for checking its transitions.

Scope of the formal claim. The Lean result is the fixed- 4×4 reachability and exact minimum round count, together with the supporting rule and mass lemmas just described. It does not formalise the paper’s largest-tile upper bound, score ceiling, longest-game bound, primed-board classification, symmetry orbit, general-dimensional claims, or optimal stochastic policy. In particular, neither failure of the naive snake strategy nor success of this certificate proves or disproves general rectangular-board attainability. Every prescribed finite spawn sequence has positive probability under the usual random rules, but the probability measure itself is not formalised here. As in any proof-assistant development, trust in the logic and kernel and review that the formal definitions express the intended game remain separate from execution of the searcher. We make no unconditional historical-priority claim for this formalisation.

6.6 Playing it

The application plays a line move by move. With planned spawns each move is exactly the step. Under honest spawns the player presses undo whenever the spawn differs from the plan and lets the game re-roll; each move is a run of geometric trials, counted in Section 8. Every position shown is a legal position of the game, and by Theorem 22 the same data serves all four corners and both directions out of each.

7 The eight spirals

The dihedral group D_4 of the square (four rotations, four reflections) acts on the cells of the board and on the four directions: a rotation by a quarter turn sends up to right, right to down, and so on; the reflection in a vertical axis swaps left and right and fixes up and down; and similarly for the rest.

32	64	8192	16384
16	128	4096	32768
8	256	2048	65536
4	512	1024	131072

bottom-right corner, chain running up

32	16	8	4
64	128	256	512
8192	4096	2048	1024
16384	32768	65536	131072

bottom-right corner, chain running left

Figure 2: The final position of the perfect game: the full chain 131072, 65536, $\dots$, 4 along the two corner snakes out of the bottom-right corner. The left board is the application's default. Every cell holds a different power of two, so the position is dead; it is simultaneously the end of the 65,533-move line, the end of the 129,333-move maximum-score line, and by Corollary 13 the unique maximum-mass position of the game. The other six spirals are its images under the symmetries of the square (Theorem 22).

Lemma 21 (equivariance). *For every $g \in D_4$, position B and direction d ,*

$$\text{slide}_{gd}(gB) = g \text{slide}_d(B),$$

with the same multiset of merged values. Consequently g maps every play to a play (transform each direction and each spawn cell) with the same length, the same score, the same spawned values, and the same number of empty cells after every slide.

Proof. A slide in direction d applies one fixed one-dimensional rule to each line parallel to d , the line read from the wall the tiles move toward. The symmetry g maps the lines parallel to d bijectively onto the lines parallel to gd and maps the wall d points to onto the wall gd points to, so it preserves the reading order of every line; the rule is the same, hence the identity, merge for merge. A spawn cell is empty before g is applied iff its image is empty after. $\square$

Theorem 22 (eight spirals). *(i) There are exactly eight corner snakes, and D_4 acts on them simply transitively: each is the image of any other under exactly one symmetry.*

(ii) If a play ends on the chain along the snake σ , then its image under $g \in D_4$ ends on the chain along $g\sigma$. Hence each shipped line yields eight plays with pairwise distinct final positions, of equal length, score and undo cost. In particular every one of the eight spiral arrangements of the full chain is reachable, each in exactly 65,533 moves (the minimum) and each with 3,925,224 points by a 129,333-move play; and the 131072 can be built in any corner in 32,781 moves with the chain leaving the corner along either side.

(iii) Every corner snake ends at the corner adjacent to its start across the side perpendicular to its first direction, so the 4 sits at a corner with the 8 beside it in the snake's last line, as Proposition 15 requires.

Proof. (i) A corner snake is determined by its start corner and the side it first runs along, that is, by a *flag* (corner, incident side); there are $4 \cdot 2 = 8$ flags and the boustrophedon rule produces

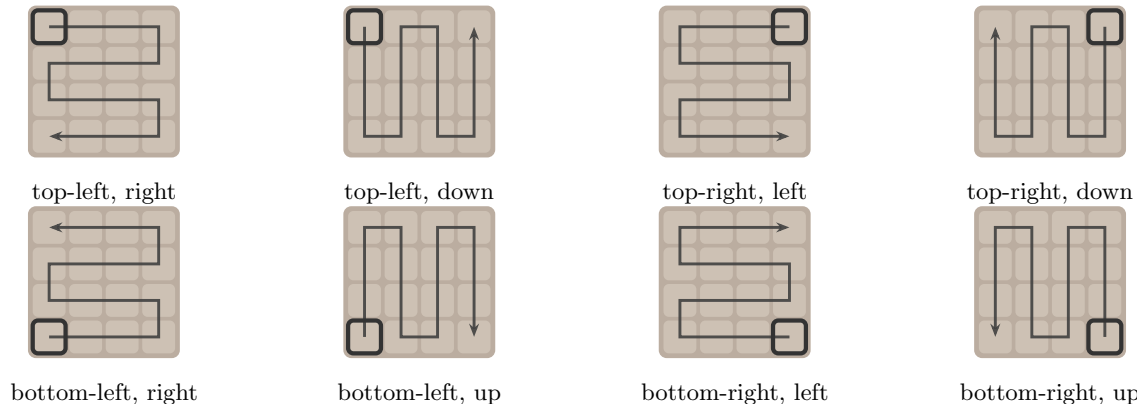


Figure 3: The eight corner snakes: one orbit of the symmetry group of the square. The marked cell is the corner that holds 131072; the arrow runs along the chain to the 4. The application exposes exactly this choice: a corner and a direction out of it.

a Hamiltonian path from each. A symmetry that fixes a flag fixes two adjacent corners and is therefore the identity, so the stabiliser of a flag is trivial and the orbit of a flag under the group of order 8 has eight elements: all of them. (ii) is Lemma 21 applied to the whole play, together with Theorem 19 for the numbers; the eight final positions differ in the corner holding 131072 or in the cell holding 65536. (iii) The snake visits the four lines parallel to its first direction in turn, so its last cell is in the line farthest from the start, at the end nearest the start's column (an even number of lines are traversed after the first); that is the corner adjacent across the perpendicular side. $\square$

Remark 23 (has this been shown before?). The snake in one corner is the standard human strategy and the standard target of heuristic players; the weight tables of Xiao's expectimax [40] and the priority map of Richardson's player [31] anchor one corner and one direction. Strategy guides describe one corner and one orientation. That the eight images are all attainable, in the same number of moves and by the same move list transformed, is an elementary consequence of the equivariance of the rules which we have not found stated in the literature; we record it because it is what makes one computed line eight perfect games, and because it is exactly the choice the application offers. Fifty-two Hamiltonian paths leave a corner of the 4×4 board, sixteen of them ending at a corner—always an adjacent one, since a Hamiltonian path on sixteen cells ends on the opposite colour of the checkerboard—and just two of the fifty-two are corner snakes. Which of the other arrangements of the chain, Hamiltonian or not, are reachable is open (Section 13); Proposition 15 rules out every arrangement with the 4 in the interior.

8 The undo button

8.1 The shipped lines

Three lines were generated once and are shipped as data: the 32,781-move line to 131072, the 65,533-move line to the full chain, and the 129,333-move maximum score line. The other three corners and the column-first spirals are the seven symmetries of the square applied to the same data (Theorem 22). A replay through the real engine checks that every slide moves, every spawn cell is empty, and the counts land exactly.

8.2 What a perfect game costs in undos

Proposition 24 (undo cost). *Under honest spawns with a re-randomising undo, the expected number of undo presses needed to play a fixed line of L moves is*

$$\sum_{t=1}^L \left(\frac{e_t}{q_{v_t}} - 1 \right),$$

where e_t is the number of empty cells after slide t and $q_2 = 0.9$, $q_4 = 0.1$. A prescribed opening—values v_a, v_b in two prescribed cells of a c -cell board—needs an expected $c(c-1)/(2q_{v_a}q_{v_b}) - 1$ restarts.

Proof. The spawn after slide t lands in the required cell with the required value with probability $p_t = q_{v_t}/e_t$, independently at each attempt; the number of attempts is geometric with mean $1/p_t$, so the undos for the move are $1/p_t - 1$ in expectation, and expectations add. The opening places its two tiles in order, uniformly in c then $c-1$ cells, and both orders produce the prescribed pair. $\square$

Two 4s in two prescribed cells of the 4×4 board happen one game in 12,000, so the line to 131072 starts with an expected 11,999 restarts. Summing the proposition over the shipped lines:

line	moves	$\mathbb{E}$ [undos in play]	$\mathbb{E}$ [opening restarts]	total
131072 (all 4s)	32,781	1,929,679	11,999	1,941,678
full chain (all 4s)	65,533	3,521,967	11,999	3,533,966
maximum score (2s)	129,333	658,984	147	659,131

Measured runs match: the browser reports 1.91–1.97 million undos for the 131072 line. The maximum-score line is four times longer but three times cheaper in undos, because a 2 is nine times as likely as a 4. By Lemma 21 the eight spirals cost the same.

9 Honest play

For the full board we implemented an expectimax player (GENIUS) in the tradition of Xiao’s [40]: rows and columns are table lookups over ranks, the heuristic scores empty cells, available merges, monotonicity and a tile-sum penalty, with a pull toward the chosen corner; depth adapts to the number of distinct tiles. Against Richardson’s [31] Evil generator, which places each new tile on the edge the last move packed against beside the largest neighbours, the search models that rule exactly instead of averaging over luck. We ported Richardson’s four players (Smart, Algorithm, Priority, Random) and the generator to the same array engine for a like-for-like comparison. Ten games each, bottom-right corner:

player	regular tiles		Evil tiles	regular, undo to escape death
GENIUS	16384 $\times$ 1, 8192 $\times$ 5, 4096 $\times$ 3, 1024 $\times$ 1; 130,131	4096 $\times$ 6, 2048 $\times$ 4; 56,850		32768 $\times$ 2 (2 games)
Smart	4096 $\times$ 3, 2048 $\times$ 6, 1024 $\times$ 1; 46,029	1024 $\times$ 4, 512 $\times$ 4, 256 $\times$ 2; 8,410		16384 $\times$ 3, 8192 $\times$ 2
Algorithm	512 $\times$ 5, 256 $\times$ 2, 128 $\times$ 3; 4,256	256/128/64; 2,011		1024 $\times$ 6, 512 $\times$ 4
Priority	512 $\times$ 1, 256 $\times$ 6, 128 $\times$ 2, 16 $\times$ 1; 2,952	128/64; 1,284		—
Random	128 $\times$ 6, 64 $\times$ 3, 32 $\times$ 1; 983	128/64/32; 508		256 $\times$ 9, 512 $\times$ 1

Semicolons separate the largest-tile histogram from the average score. The last column is the undo button used as a human uses it: only to escape game over, each death unwinding a doubling number of moves, giving up after forty deaths without a new best score.

question, standard 4×4 game	status
optimal move in every position; optimal expected score	open; 4×3 solved [15]; boards up to nine cells and the cube solved here (Table 2)
a strategy reaching 2048 with high probability	yes, probability ≥ 0.99969 [37]
a tile reachable with certainty	256 [37]; the optimal probability of each larger tile is open
largest tile	131072: bound folklore and [7], Corollary 12; reachable: the certificate of Section 6.4 (claimed for all boards by [7])
fewest moves to 2048	519, exact: [21] and Theorem 4
fewest moves to 131072	32,781, exact and kernel-checked in Lean (Theorem 20); primed-board classification proved separately (Lemma 5)
fewest moves to 131072, two initial 2s	32,782, exact and kernel-checked in Lean (Theorem 20)
maximum score	between 3,925,224 and 3,932,100 (Theorems 9, 19); conjectured 3,932,100
longest game	between 129,333 and 131,052 (Theorems 14, 19)
the full chain 131072, $\dots$, 4	reachable, in exactly 65,533 moves at the fewest (Corollary 6); at least 320 arrangements have minimum-length certificates (Section 11), with the largest tile in every boundary cell; every reachable arrangement has its 4 on the boundary beside the 8 (Proposition 15); which other arrangements are reachable is open
probability that an honest game reaches 131072	open; between $10^{-48,607}$ and $(1 + 10^{-65535})/11$ by Section 11; the exact optimum is open
number of reachable positions	open; more than $1.3 \cdot 10^{12}$ [19]

Table 3: What is known about the standard game, September 2026.

10 Where the 4×4 game stands

Kaneko and Yamashita’s strong solution of 4×3 [15] had to enumerate $1.15 \cdot 10^{12}$ positions for twelve cells. For the sixteen cells of the standard board nobody knows the count: Lees-Miller’s combinatorial bound is about $4.4 \cdot 10^{16}$ positions for the game up to the 2048 tile and is known to be a large over-estimate [20], and his enumeration ran through $1.3 \cdot 10^{12}$ positions in a month without finishing [19]. *Strongly* solving 4×4 —the optimal move in every reachable position and hence the optimal expected score and the optimal probability of every tile—is out of reach of this paper and, as far as we know, of any current computation; the layered pass of Section 5.5 is the right shape for it (it is Kaneko and Yamashita’s), and a central obstacle is the state-space size. What the previous sections settle is the other half of “perfect 2048”, the possible-play half. Table 3 lists both halves.

Reaching the largest tile. The upper bound 2^{c+1} is elementary (Corollary 12); the question has always been attainability under the actual slide rules. Rodgers and Levine [34] called 131072 attainable “in theory” and the video evidence for it suspect; Lees-Miller [20] obtained it from tile counting alone and left the geometric question open. Das and Paul [7] state attainability for every $n_1 \times n_2$ board as their Theorem 1. Their argument prescribes the spawns and proceeds by induction on the board, performing the construction for an $n_1 \times n_2$ board inside an $(n_1 + 1) \times n_2$ board and then using the extra cells; but a slide moves every line of the board, so a fixed-coordinate sub-board is not isolated from the rest. Proposition 29 supplies a moving-window invariant that validates embedding itself. The remaining use of the additional cells for further doublings still needs a complete strategy and invariant; we do not claim that general construction here. What is established: on the enumerated boards up to ten cells, the cube included, by exhaustive enumeration (Table 1); on 4×3 by Kaneko and Yamashita’s enumeration, which reaches $8192 = 2^{13}$;

and on 4×4 by the certificate of Section 6.4, whose legality and minimum length are kernel-checked in Section 6.5. A uniform multidimensional construction with a proven invariant remains an unresolved verification task in this work; the line family is handled in Section 11; the failed naive strategy alone does not establish that the general theorem is false or historically unsolved: a snake counter that spawns at the far end of the frontier row and slides toward its head fails at the first turn of every board we tried, and our 4×4 line was found by search, not by a rule.

Predictions for 4×3 . Kaneko and Yamashita’s pass carries, per position, what the optimal policy needs. Adding a two-byte “fewest spawned 4s” field and a “fewest moves” field would decide the possible-play half of 4×3 with additional per-state storage, and this paper predicts the answers: the largest tile 8192 in $M(13) = 2,057$ moves and not fewer; the full chain $8192, \dots, 4$ in $2^{12} - 3 = 4,093$ moves at the fewest and with exactly twelve spawned 4s at the fewest; a maximum score of $\Phi_{12} - 48 = 180,180$; a longest game of $2^{13} - 16 = 8,176$ moves; and $4,4 \rightarrow 8$ from a full board as the last move of every full-chain play. Every one of these predictions has held on every board we could solve.

Three dimensions. Das and Paul’s game on a $2 \times 2 \times 2$ cube has eight cells, six directions and a symmetry group of order 48. Our solver handles boxes of any dimension (a slide is the same line rule along any axis), and the cube takes fourteen seconds: 702,581 positions up to symmetry, 32,187,234 raw. Every bound of this paper is attained on it: the largest tile $512 = 2^9$ is reached in exactly $M(9) = 133$ moves, the maximum score is $\Phi_8 - 32 = 7,140$, the longest game is $2^9 - 12 = 500$ moves, the full chain needs $253 = 2^8 - 3$ moves and eight spawned 4s, and all 9,216 chain-producing moves are $4 + 4 \rightarrow 8$ from a full board. In the honest game the cube is a better board than the 2×4 strip with the same eight cells: an optimal expected score of 2,953.42 against 2,641.87, a 256 tile with probability 94.04% against 86.98%, and a 512 with 6.32% against 2.46%. Six directions beat four.

11 Further extremal deductions

The formal witness also supports stronger questions than the first existential optimum. We now distinguish the minimum for a *fixed* opening, the least possible mass when the target is reached, and the largest score among the fastest games. Separately, binary mass barriers yield policy-independent stochastic bounds and a useful restriction of the unresolved maximum-score search. Only the results explicitly marked as Lean-checked below are included in the extended formal development.

11.1 Joint time–mass optima for every opening

Theorem 25 (Every-opening joint optimum; Lean-checked). *Let A be any ordinary two-tile opening of standard 4×4 2048, and put*

$$m(A) = \begin{cases} 32,781, & \text{both opening tiles are 4s,} \\ 32,782, & \text{otherwise.} \end{cases}$$

There is a legal play from A of exactly $m(A)$ rounds ending with a 131072 tile and total mass 131102. Every legal play from A reaching that tile has at least $m(A)$ rounds and endpoint mass at least 131102.

The quantifiers are stronger than in Theorem 20: *for every opening there exists a continuation*. There are 480 labeled openings: 120 with two 4s, 240 mixed, and 120 with two 2s. They form 75 square-symmetry classes. A finite bridge search finds, for each opening, at most three preparatory rounds joining a symmetry of one of the first four states of the original certificate. The length of the remaining suffix is accounted for exactly, rather than adding a blanket three-round overhead. A separately proved symmetry transport, every bridge, and exhaustive coverage of the ordinary-opening definition are checked in Lean. The new theorem is

`Game2048.standard2048.all_openings_exact.`

The constructive half has no axiom dependencies; the combined theorem uses only `propext` and `Quot.sound`.

Mass lower bound and construction. Continue to write $H_k(B)$ for the mass in tiles strictly greater than k . A round $B \rightarrow B'$ with $k \geq 2$ satisfies

$$H_{2k}(B') \leq H_k(B), \quad \text{mass}(B') \geq \text{mass}(B) + 2.$$

Thus the mass outside the increasing threshold grows by at least two per round. Induction over any final r rounds gives

$$H_{2r+1}(B') + 2r \leq \text{mass}(B').$$

With $r = 15$, a 131072 tile contributes to H_{65536} , so the endpoint mass is at least $131072 + 30 = 131102$. The existing move lower bound ensures that these fifteen rounds exist. This argument is proved from the executable rules in `ResidualMass.lean`.

A new fifteen-round cascade uses only 2-spawns after the original 32,766-round primed prefix. It attains both the mass bound and the move bound. The bridge construction transports this cleaner suffix to every opening. For an opening of mass at most 6, the same threshold inequality used earlier gives $131072 \leq 6 + 4(T - 15)$ and hence $T \geq 32,782$; mass 8 gives 32,781. $\square$

The new minimum-mass witness uses 32,768 spawned 4s and fifteen spawned 2s, opening included, and scores 1,966,148. Its score is independently replayed, not asserted by the Lean mass theorem.

11.2 Maximum score among the fastest games

Theorem 26 (Fastest-game score optimum). *Among ordinary 4×4 plays reaching 131072 in exactly 32,781 rounds, the largest score at that time is 1,966,216.*

Proof. The minimum-time equality argument forces all 131072 mass at round 32,766 to contribute to the target. All earlier births are 4s, and none of the final fifteen spawns can contribute to its ancestry. The target's merge tree therefore contributes

$$(17 - 1)131072 - 4 \cdot 32768 = 1,966,080.$$

Only the other fourteen births before the last spawn can contribute additional merge score; the last spawn itself is unscored. If y of these fourteen births are 4s, their total mass is $28 + 2y$. Repeatedly merging equal values in a geometry-free multiset increases its potential and ends at the binary expansion of its mass. Write $\Phi_{\text{bin}}(s)$ for that expansion's potential. The additional score is at most

$$\Phi_{\text{bin}}(28 + 2y) - 4y, \quad 0 \leq y \leq 14.$$

For these fifteen values of y , the bounds are respectively

$$68, 64, 120, 116, 116, 112, 120, 116, \\ 116, 112, 136, 132, 132, 128, 136.$$

The maximum is 136. A second new fifteen-round cascade uses all 4s and ends with the target together with junk tiles 32, 16, 8, 4, attaining $1,966,080 + 136$. Both independent rule replayers accept the complete trajectory in `witness/131072-fold-4.txt`. $\square$

This is an optimum subject to the shortest-game constraint, not the unrestricted 3,932,100 score conjecture. Replacing its last spawn by a 2 preserves the score. In particular the original minimum-length certificate was not optimized for either endpoint mass or score:

trajectory	rounds	endpoint mass	score
original minimum-length witness	32,781	131,116	1,966,092
new minimum-mass continuation	32,781	131,102	1,966,148
new fastest-game score optimum	32,781	131,132	1,966,216

A second Pareto endpoint: one spawned 4. The gateway theorem below forces at least one 4-birth. Appending a new all-2 cascade to the first gateway of the score-oriented witness gives 131072 in exactly 65,548 rounds with just one 4-birth, opening included, mass 131102, and score 2,097,216. This is the fewest rounds possible under that resource constraint, because $131072 \leq 4 + 2(T - 15) + 2$ forces $T \geq 65,548$. It also has the largest score among minimum-mass target endpoints: the non-target mass 30 has binary potential at most 68, and the required 4-birth costs at least four points, giving $\Phi(131072) + 68 - 4 = 2,097,216$. Both existing replayers check `witness/131072-one-four.txt`. This low-4 prefix is independently replayed, not included in the extended Lean certificate.

11.3 Unavoidable mass gateways

Theorem 27 (Gateway theorem). *Let $c \geq 3$, $C = 2^{c+2}$, and $G_j = C - 2^{j+2}$ for $j = 0, \dots, c - 1$. Every ordinary play reaching mass at least G_j visits mass exactly G_j , entering it by a 4-spawn. Its tile multiset there is*

$$(\{4, 8, \dots, 2^{c+1}\} \setminus \{2^{j+2}\}) \cup \{4\},$$

where the final union adds one tile, including a second 4 when $j > 0$.

Proof. The binary expansion of $G_j - 2$ has c nonzero bits. Any representation by at most c powers of two is consequently that expansion itself: the board is full, its values are distinct, and it is dead. A first crossing of G_j cannot depart from $G_j - 2$, nor can it jump over G_j . It must go from $G_j - 4$ to G_j by spawning a 4. Immediately before that spawn there are at most $c - 1$ tiles, because one cell must be vacant, and $G_j - 4$ has binary popcount $c - 1$. Its afterstate multiset is therefore the binary expansion, which fixes the claimed post-spawn multiset. For $c \geq 3$ every ordinary opening lies below the first gateway. $\square$

For 4×4 the gateways are, in order,

$$131072, 196608, 229376, 245760, 253952, 258048, \\ 260096, 261120, 261632, 261888, 262016, 262080, \\ 262112, 262128, 262136, 262140.$$

Thus *every* route to 131072 passes through the primed multiset, not merely the fastest routes. Every full-chain play needs a 4-spawn at each of the sixteen gateway entries. The argument is independent of any snake invariant or merge-tree scheduling theorem.

11.4 Stochastic consequences of the gateways

Let each birth independently be 4 with probability p and 2 with probability $q = 1 - p$. In units of 2 the birth-mass process is a walk with increments 1 and 2. Its probability of ever hitting $n \geq 0$ is

$$u_n = \frac{1 - (-p)^{n+1}}{1 + p},$$

as follows from $u_0 = 1$, $u_1 = q$, and $u_n = qu_{n-1} + pu_{n-2}$. A gateway at even distance D can be crossed only by hitting $D - 2$ and next adding 2, not by hitting the preceding dead layer. Its mass-only success probability is

$$b(D) = pu_{D-2} = \frac{p}{1 + p}(1 + p^{D-1}).$$

Extending the independent birth sequence beyond an actual game's stopping time gives an unrestricted walk containing every successful game as an event. Discarding geometry and other failure conditions can only increase the success probability. This coupling proves the following bounds for every history-dependent player policy.

Corollary 28 (Policy-independent probability bounds). *On a c -cell board with $c \geq 3$,*

$$\Pr(\text{reach } 2^{c+1}) \leq \frac{p}{1 + p}(1 + p^{2^c-1}),$$

and

$$\Pr(\text{reach the full chain}) \leq \left(\frac{p}{1 + p}\right)^c \prod_{k=1}^c (1 + p^{2^k-1}).$$

Proof. The first event requires crossing the first gateway, whose unit-mass distance from zero is 2^c . Successive distances for the full chain are $2^c, 2^{c-1}, \dots, 2$; future increments are independent of the history upon reaching each gateway. Multiply the corresponding $b(D)$. The two initial tiles are the first two births, both before the first barrier for $c \geq 3$. $\square$

For standard 4×4 play, with $p = 1/10$, these become

$$\Pr(131072) \leq \frac{1 + 10^{-65535}}{11} < 0.090909091$$

and a full-chain bound of approximately $2.3963146538 \cdot 10^{-17}$. The exact expression in the corollary is the upper bound; a rounded decimal must not silently replace it. In particular, the positive correction in the first bound cannot be dropped to claim $\leq 1/11$.

There is also a better constructive lower bound than the original minimum-length transcript provides. The one-4 witness first reaches 131072 at round 65,548. Its exact rational probability, including the prescribed two-cell opening and every empty-cell spawn choice, is at least 10^{-48607} . The computation factors that rational number and proves the comparison by integer arithmetic. Hence

$$10^{-48607} \leq \sup_{\pi} \Pr_{\pi}(131072) \leq \frac{1 + 10^{-65535}}{11}.$$

This interval is wide and is not an estimate of an optimal agent’s win rate. The new lower bound improves the valid 10^{-57048} bound from the original shortest witness by 8441 powers of ten. The earlier printed 10^{-57047} rounded that witness’s probability in the wrong direction; its logarithm is approximately -57047.3719 .

11.5 The value word of a ceiling-score game

The score equality case has two post-spawn alternatives: the full chain with c spawned 4s, or the same chain with its last 4 replaced by a 2 and $c - 1$ spawned 4s. The last slot in the score inequality contributes zero in either case. Every earlier slot must be saturated. Changing the final spawn from 4 to 2 changes no merge score.

The gateway theorem forces all $c - 1$ nonfinal 4-births in either ceiling-score play. There can be no other pre-final 4-births, so all remaining births, including the opening, are 2s. Therefore the spawn *values* are fixed up to an arbitrary final 2 or 4: the unresolved search need choose only directions and empty cells, not additional 2-versus-4 decisions. There are $2^{c+1} - 2c - 2$ forced 2-births and $c - 1$ forced 4-births before the final arbitrary birth. Consequently,

$$\Pr(\text{attain } \Phi_c - 4c) \leq q^{2^{c+1}-2c-2} p^{c-1}.$$

For 4×4 this is $0.9^{131038} 0.1^{15} \approx 1.0714006617 \cdot 10^{-6011}$. Ripà already gave this value-word probability and the numerical score and length formulas in 2014 [32, 33]; we do not claim them as new. This necessary-event bound does not assume that the score ceiling is attainable. Our bounded fixed-value-word search did not complete a ceiling-score witness and supplies no impossibility result.

11.6 New full-chain arrangements

An exhaustive 64-round suffix search from round 65,469 of the existing all-4 full-chain witness gives 32 distinct terminal D_4 orbits. Each complete chain has sixteen distinct values, hence trivial stabilizer and eight labeled boards per orbit. A separate 70-round checkpoint bridge supplies eight further noncorner-largest-tile orbits. Deduplicating the union gives 40 orbits and at least **320** arrangements with 65,533-round witnesses, all optimal by Corollary 6. It does not classify all reachable endpoints: only continuations of that fixed prefix were enumerated.

In particular, the question whether the 4 can be diagonally opposite 131072 has a positive answer. A sixteen-round replacement tail gives

$$\begin{array}{cccc} 4 & 8 & 16 & 32 \\ 64 & 128 & 256 & 512 \\ 8192 & 4096 & 2048 & 1024 \\ 16384 & 32768 & 65536 & 131072 \end{array} \cdot$$

The full transcript is `witness/full-chain-opposite.txt`. The off-corner question also has a positive answer. A 70-round legal bridge between rounds 49,166 and 49,236, followed by a reflected suffix with the two distinct inert largest labels exchanged, produces

$$\begin{array}{cccc} 4 & 8 & 16 & 32 \\ 512 & 256 & 128 & 64 \\ 1024 & 2048 & 4096 & 8192 \\ 65536 & 131072 & 32768 & 16384 \end{array} \cdot$$

The explicit transcript `witness/full-chain-noncorner.txt` is replayed in its entirety; no arbitrary label exchange is permitted as an operation in the actual game. The exchange describes how the candidate suffix was found, not how its legality is justified. Symmetries put the largest tile in every boundary cell. Interior placement in a full-chain endpoint and a complete classification remain unresolved. Forty-eight representative histories (including overlaps) are independently replayed by both rule implementations, and the endpoint union is explicitly deduplicated. Reports are retained under `research/results/`.

11.7 A uniform one-dimensional family and a valid embedding

On a line with $c \geq 2$ cells, start with 4s at opposite ends and repeatedly slide left and spawn a 4 at the far right. Nonzero values remain nonincreasing. A non-full board moves because the last birth is at the right edge; a full board with a duplicate has an adjacent merge. The only full, distinct multiset with values at least 4 under the tile ceiling is the full chain. Thus it is reached in $2^c - 3$ rounds. The first primed configuration folds by one carry per round, giving the largest tile in $2^{c-1} + c - 3$ rounds, exactly the general lower bound.

For maximum score, start with opposite-end 2s and spawn a 2 except when the after-slide mass is $G_j - 4$, when a 4 is required. Those exceptional afterstates contain no 2s, so order remains nonincreasing. A full distinct board containing a 2 must be at a dead layer $G_j - 2$; the rule skips every such layer. It ends on the full chain with exactly c 4-births, attaining both $\Phi_c - 4c$ points and $2^{c+1} - 4 - c$ rounds. This is an infinite-family construction, not merely the finite tests (lengths 2 through 16) supplied with the paper. It is not a construction for arbitrary rectangles.

Proposition 29 (Floating-window embedding). *A legal play on a box of side lengths n_i embeds in any box of side lengths $N_i \geq n_i$, preserving its directions, spawn values, number of rounds, and merge score, with translated spawn coordinates.*

Proof. Keep all tiles in a translated $n_1 \times \cdots \times n_d$ window. After sliding along axis j , set that window's offset on axis j to 0 toward the low wall or $N_j - n_j$ toward the high wall. Other offsets are unchanged. Empty padding changes no ordered nonzero line sequence and hence no merges; compression gives the translated small-board result. The window moves in the same direction as the slide and cannot cancel a nontrivial displacement. A merge likewise stays nontrivial. Translate the prescribed spawn into its still-empty window cell. Induction over the play proves the claim. $\square$

Thus a fixed-coordinate sub-board is not isolated, but a moving-window invariant does justify embedding. This repairs that part of the concern raised about Das and Paul's induction; it does not prove their further claim that every new cell buys an additional doubling. Independent lifts of the entire shortest witness to 5×5 , 5×7 , and 4×8 check the invariant move by move.

11.8 A computable upper bound on optimal expected score

Use mass units of 2, put $H = 2^{c+1} - 2$, and let $F(m)$ be the potential of the binary expansion of actual mass $2m$. For $c \geq 3$, there is a Bellman relaxation with only $H + 1$ scalar mass states. Stop the increment walk at its first mass with c binary ones. Define

$$W(m) = \begin{cases} F(m), & s_2(m) = c, \\ qW(m+1) + p(W(m+2) - 4), & s_2(m) < c, \end{cases} \quad 0 \leq m \leq H.$$

All successors used in the second case are at most H . This backward recurrence exactly computes the expected terminal binary potential minus four times the number of future 4-spawns in the relaxation.

To justify it as an upper bound, at spawn index n let m_n be the mass and f_n the number of 4-spawns, and put $R_n = F(m_n) - 4f_n$. The actual score is at most R_n . Moreover R_n is nondecreasing: a 2-spawn changes it by $F(m+1) - F(m) \geq 0$, and a 4-spawn changes it by $F(m+2) - F(m) - 4 \geq 0$. The real game cannot continue beyond the walk's first terminal barrier. Extending the pre-sampled walk after an earlier geometric loss therefore still gives an upper bound. In particular, for an ordinary random opening,

$$\sup_{\pi} \mathbb{E}_{\pi}[\text{score}] \leq W(0).$$

From an already given board B of mass m in units of 2, the analogous upper bound on future score is $W(m) - \Phi(B)$. A second recurrence $J(m) = 1 + qJ(m+1) + pJ(m+2)$, with zero boundary values, bounds the expected game length by $J(0) - 2$.

For $c = 16$, outward-rounded decimal evaluation gives

$$\begin{aligned} \sup_{\pi} \mathbb{E}_{\pi}[\text{score}] &< 1,915,854.958, \\ \sup_{\pi} \mathbb{E}_{\pi}[T] &< 62,412.330. \end{aligned}$$

For comparison the four-cell score relaxation is $69.814912\dots$, above the exact geometric 2×2 value in Table 2. These are bounds, not strong solutions. The script `research/mass_interval.py` uses fifty-digit arithmetic rounded outwards at every operation and records enclosing intervals; a separate forward computation checks the values numerically.

11.9 Near-optimum rigidity and deadline probabilities

If a 4×4 play reaches 131072 at round $T = 32781 + d$, inspect its prefix ending fifteen moves earlier. The threshold-mass lemma requires at least 131072 mass there. If z is the number of 2-spawns in that prefix, excluding the opening, and M_0 is the opening mass, the ledger therefore gives

$$2z + (8 - M_0) \leq 4d.$$

The left side also counts the opening 2s, at a charge of two mass units each. Thus among the first $32768 + d$ spawns, including the opening, there can be at most $2d$ twos. The deterministic prefix inequality, its general-target version, and the zero-slack consequence are kernel-checked in `Game2048/DeadlineSlack.lean`.

Pre-sampling values again gives a bound for *any* policy reaching the tile by the deadline:

$$\Pr_{\pi}(T_{131072} \leq 32781 + d) \leq \sum_{z=0}^{\min(2d, 32768+d)} \binom{32768+d}{z} (0.9)^z (0.1)^{32768+d-z}.$$

For an earlier hit the required mass has already been acquired by the specified prefix, so the same necessary event applies. At $d = 0$ this is 10^{-32768} . For deadlines 40000, 50000, and 55000, the base-ten logarithms of the upper bounds are approximately -14854.03 , -3668.61 , and -909.55 . These are spawn-value bounds; requiring suitable locations can only decrease the success probability. The stochastic corollary is not separately formalised in Lean.

Reproduction and scope. The scripts in `research/` generate the short bridges, clean cascades, suffix variants, exact probability bounds, and line tests. `bash research/finish.sh` runs the extended checks after the original formal development is rebuilt. The symmetry, every-opening, endpoint-mass, and deterministic deadline-slack theorems are kernel-checked. The other new deductions have written proofs and executable checks, and are not thereby claimed to have been completely formalized in Lean.

12 Structural reachability and reproducible state-space bounds

This section concerns boards after the mandatory spawn, including every ordinary two-tile opening. Accumulated score and random-generator state are not part of the counted object. We distinguish labeled boards from orbits under the eight square symmetries. A run stopped at target T counts boards whose maximum tile is strictly below T ; individual losing boards remain distinct and there are no artificial win or loss nodes. The following arguments are human proofs with executable checks, not extensions of the Lean formalization in Section 6.5.

12.1 A direct rank invariant

For a c -cell board B , let $N_q(B)$ be the number of tiles with value at least 2^q .

Theorem 30 (rank envelope). *For every reachable board with $c \geq 2$ cells,*

$$N_q(B) \leq c + 2 - q \quad (2 \leq q \leq c + 2).$$

Equivalently, decreasing nonzero ranks satisfy $r_p \leq c + 2 - p$. Consequently no tile exceeds 2^{c+1} and

$$\text{mass}(B) \leq 2^{c+2} - 4.$$

Equality in the mass bound requires the unique multiset $\{2^{c+1}, 2^c, \dots, 4\}$.

Proof. The opening satisfies the inequalities; the case $q = 2$ is the cell capacity. Suppose the bounds hold before a slide and fix $q \geq 3$. Write $H = N_q(B)$ and let j be the number of disjoint merges of two rank- $(q - 1)$ tiles. Only these merges can increase N_q . If $j = 0$, its old bound persists. Otherwise,

$$H + 2j \leq N_{q-1}(B) \leq c + 3 - q, \quad N_q(\text{after}) \leq H + j \leq c + 2 - q.$$

A spawn has rank at most 2, so the induction closes, including parallel merges. No rank $c + 2$ can occur, and larger ranks cannot arise without first producing it. Sorting the threshold inequalities yields the rank form. Sum its slotwise tile bounds to obtain the mass bound and its unique equality multiset. $\square$

This proof is independent of the earlier tight-moment argument. On 4×4 it gives $\text{mass}(B) \leq 262140$ and a tile bound of 131072; attainment still relies on the separately checked witnesses.

12.2 Recent-spawn constraints

Let $L_q(B)$ be the total mass in tiles strictly below 2^q .

Lemma 31 (recent-spawn mass). *If B contains a tile of rank at least $q \geq 3$, then*

$$L_q(B) \geq 2(q - 2).$$

Proof. A spawn made j rounds before B has experienced at most j subsequent merges along its ancestry, so its current descendant has rank at most $j + 2$. All of the latest $q - 2$ spawns therefore have descendants below rank q , and their disjoint ancestral masses sum to at least $2(q - 2)$. Reaching rank q from an ordinary opening requires at least $q - 2$ rounds, so these spawns exist. $\square$

A stronger necessary test records forced recent 4-spawns. Let n_r count rank- r tiles, let K be the maximum rank, and let $E(B)$ be the occupied cells whose deletion leaves occupancy compact toward some wall. The last spawn must be an eligible 2 or 4. Set $f = 0$ when an eligible 2 exists, otherwise $f = 1$ when an eligible 4 exists, and reject if neither does. For $r = 3, \dots, K$, set

$$j = r - 2, \quad U_r = \sum_{s < r} n_s 2^{s-1}.$$

Whenever $U_r \leq j + f$, require $n_r > 0$ and increase f by one; otherwise leave f unchanged.

Lemma 32 (causal supply). *Every noninitial reachable board passes this test.*

Proof. Let F_j be the actual number of 4-spawns among the latest j spawns. Inductively $f \leq F_j$. These j spawns have descendants below rank $r = j + 2$ and occupy $2j + 2F_j$ of the available lower-rank mass $2U_r$. If $U_r \leq j + f$, there is no remaining lower-rank mass for the next older spawn to be a 2: its descendant would still be below rank r . It must be a 4, and there must be a rank- r tile to receive mass that cannot fit below r . Increasing f preserves the lower bound. When the inequality fails, leaving f unchanged is a relaxation. A rejection can occur only at a missing rank $r \leq K$, whence $K \geq r + 1 = j + 3$; at least $j + 1$ rounds have then occurred, and the older spawn used in the contradiction exists. No pre-opening spawn is assumed. $\square$

For example, the inventory $\{128, 8, 2\}$ passes Lemma 31 but fails Lemma 32. The last spawn must be the 2; the preceding spawn must be a 4 to contribute to the 8 with only one subsequent merge available. The latest five spawns would then carry at least 12 mass below 128, but only 10 is present.

12.3 Exact inverse slides

Theorem 33 (inverse line rule). *Let $(y_1, \dots, y_m, 0, \dots, 0)$ be a compact target line of length ℓ , with y_i a power of two at least 2. Choose $s_i \in \{0, 1\}$. Use the input block $[y_i]$ if $s_i = 0$, or $[y_i/2, y_i/2]$ if $s_i = 1$, the latter allowed only for $y_i \geq 4$. These choices yield exactly all preimages provided*

$$m + \sum_i s_i \leq \ell, \quad s_i = 0 \implies y_i \neq y_{i+1}/2^{s_{i+1}} \quad (i < m).$$

Concatenate the blocks and insert zeros in any remaining positions.

Proof. Every output consumes either one input tile or two equal input tiles, which gives the two block forms. A pair is consumed and cannot merge onward in the same slide. A singleton must not equal the next block's first tile, or the greedy scan would merge across the proposed boundary. These conditions are necessary and make the scan consume exactly the specified blocks, proving sufficiency. Inserting zeros preserves the nonzero order. $\square$

For a four-cell line the characterization is particularly simple. A nonempty partial compact output is in the image exactly when its nonzero sequence has no adjacent pair 2, 2; a full output is in the image exactly when every adjacent pair differs. For unequal neighbors, use a shifted sequence as a preimage. For a partial output a, a, b split the first a ; for a, b, b split the last b ; for a, a, a split the middle a . The repeated value is at least 4 and Theorem 33 verifies every case. Full image lines and empty lines have only identity preimages; every nonempty partial image line has a nonidentity preimage.

With $q = 17$ allowed positive ranks, the image size within that rank range is

$$1 + q + (q^2 - 1) + (q^3 - 2q + 1) + q(q - 1)^3 = 74,818.$$

The checker exhausts all $18^4 = 104,976$ inputs and handles synthetic rank-18 outputs separately. Board predecessors are obtained by deleting each proposed last 2/4, choosing a common direction, and combining line preimages, retaining nonidentity slides and post-spawn predecessors. This describes immediate predecessors, not their reachability.

12.4 A finite characterization and a numerical upper bound

Let I be the ordinary openings, R the reachable boards, and U a proved finite superset of R . With exact successors define

$$F_0 = U, \quad F_{k+1} = I \cup (U \cap \text{Succ}(F_k)).$$

Proposition 34 (rooted predecessor filtering). *The sequence decreases, contains R at every stage, and stabilizes at exactly R . For the 4×4 rank-bounded universe of mass at least 4, $F_{131069} = R$.*

Proof. Monotonicity and $F_1 \subseteq U$ give descent. Reachable boards survive because they are openings or have reachable predecessors. At a fixed point every noninitial member has a predecessor in the same set. Following predecessors decreases mass by 2 or 4, so it must end at an opening. Every fixed-point member is therefore reachable. A chain in the stated universe has at most $(262140 - 4)/2 = 131068$ edges; unrolling the recurrence for 131069 stages leaves no orphan predecessor chain. $\square$

We count a cheaper superset, using Theorem 30, deletion-based occupancy compaction, and Lemma 32. For each symmetry, occupied cells form cycles of its permutation, and invariant tile values are constant on each cycle. Group cycles by length and eligibility, choose low values 2 and 4, and assign ranks 3 through 17 with a finite integer recurrence. At rank r , at most $18 - r$ unassigned occupied *cells* may remain. Binomial factors choose the cycles; the running lower-rank mass and f implement the causal test. Saturation at 32 units of value 2 is safe because every comparison threshold is below 32.

The fixed counts for the identity, quarter-turn, half-turn, three-quarter turn, vertical reflection, horizontal reflection, and the two diagonals are

$$\begin{aligned} &22,851,583,961,907,351,128; \quad 9,616; \quad 1,074,108,288; \quad 9,616; \\ &1,296,452,906; \quad 1,296,452,906; \quad 411,457,895,734; \quad 411,457,895,734. \end{aligned}$$

Of the 480 labeled openings, 24 lie outside the compaction set. Their fixed-count corrections are $(24, 0, 4, 0, 4, 4, 6, 6)$. Burnside's lemma gives

$$\begin{aligned} N_{\text{labeled}} &\leq 22,851,583,961,907,351,152, \\ N_{/D_4} &\leq 2,856,448,098,561,271,997. \end{aligned}$$

Separate Python-integer and JavaScript-BigInt implementations reproduce all eight fixed counts and all opening corrections. These numbers are exact sizes of necessary-condition supersets, not estimates or exact counts of the reachable set. The exact inverse-slide value restrictions are *not* included in this numerical bound.

12.5 Executed checks and reproducibility

Board and stopping rule	Symmetry classes	Labeled boards
2×2 , continued	110	662
2×3 , continued	21,752	85,844
3×3 , continued	48,713,519	388,921,077
4×4 , stop at 8	84,660	675,154
4×4 , stop at 16	23,483,970	187,809,874
4×4 , mass at most 24	67,104	535,148

These are completed regression computations, not new record claims. The separate raw-board Python search agrees with the entire symmetry expansion of the C++ dumps for 2×2 and 2×3 . On 2×2 the inverse relation is checked for all 1,296 syntactic target boards through 32. Rooted filtering gives 896,686,666,662 before stabilization. The 3×3 benchmark is reproduced under the published convention; mass-layer enumeration itself already appears in Lees-Miller’s 2017 work [19], preceding the larger 4×3 computation [15].

A 24-round 2×2 certificate ends on

$$\begin{pmatrix} 0 & 0 \\ 2 & 2 \end{pmatrix} \longrightarrow \begin{pmatrix} 2 & 8 \\ 16 & 32 \end{pmatrix}, \quad \text{score} = 180.$$

It attains the ceiling without a final 4, demonstrating why both equality cases in Theorem 9 are required. Changing a final spawn changes no merge and hence no score. The complete replay records every move, spawn, intermediate board, and cumulative score.

Run `bash research/state-space/run.sh --large` from the repository root. Each invocation uses a fresh directory, rebuilds the executable, checks every reported aggregate against its layer table, compares both bound implementations, and tests corrupt state dumps and invalid inputs. The manifest records only executed cases, source hashes, tool versions, exit codes, and output hashes. These hashes establish byte identity, not completeness; the independent set and closure checks supply the latter for the tested small boards. No full 4×4 cardinality, target-64/128 completion, or attainment of the 4×4 score ceiling is claimed.

13 Open problems

1. Strongly solving 4×4 (Table 3): the optimal expected score and the optimal probability of each tile, in particular of 131072, whose policy-independent probability bounds are sharpened in Section 11.
2. Conjecture 16: a 4×4 play with sixteen spawned 4s, equivalently a 131,052-move game.
3. The maximum score and longest game of 4×3 (predicted 180,180 and 8,176), within reach of Kaneko and Yamashita’s enumeration; the honest game of 2×5 and of the $2 \times 2 \times 3$ box.
4. A fully specified, independently checked uniform construction for Das and Paul’s claimed Theorem 1, that 2^{c+1} is reachable on every rectangular board. Our inability to complete their induction, and failure of one snake algorithm, are not disproofs of their claim. A separate

question is whether Theorems 4, 9 and 14 are tight on *every* board and in every dimension. The $2 \times 2 \times 3$ box (twelve cells) is the next three-dimensional case, at the scale of 4×3 .

5. Which arrangements of the full chain are reachable on 4×4 ? At least 320 arrangements have optimal-length certificates, including every boundary location of 131072 and the opposite-corner placement of the 4 (Section 11). Can the 131072 occupy an interior cell of a full chain? A complete classification remains open.
6. The boards passed at move $M(n) - (n - 2)$ by the fastest plays to 2^n when 2^n is not the top tile (Lemma 5 fixes them only for $n = c + 1$).
7. Minimum undos: the expected undo count above is for the shipped lines; the line minimising expected undos (which would prefer fewer empty cells for a fixed prescribed spawn value) is a different optimisation.

Reproducibility

Manuscript and artifact availability. The manuscript, plaintext witnesses, source code, and verification records are available in the public repository <https://github.com/DomTheDeveloper/2048-undo>, on its `gh-pages` branch. The submission source bundle and its build manifest are retained under `submission/`. All results are reported with their verification scope; posting this preprint does not constitute journal acceptance or external peer review.

Research expansion. Run `bash research/finish.sh` to rebuild the original certificate, the every-opening and minimum-mass theorems, the deadline-slack lemmas, and the finite calculations and new arrangement certificates. The optional `--computations-only` mode explicitly makes no fresh Lean-checking claim. `research/audit/final-verification.json` records the checked source hashes and the exact theorem dependencies. The multi-billion-state 2×5 enumeration is not rerun by these checks.

Formal proof. Run `bash lean-kernel/check.sh` from the repository root with Python 3 and the pinned Lean 4.19.0 toolchain installed. The build regenerates all transition proofs from the plaintext witness, checks the universal lower bounds and both exact-optimum theorems, and writes `lean-kernel/verification.json` after successful checking. The sources, theorem identifiers, trust boundary, and completed core audit are documented in Section 6.5 and in `lean-kernel/README.md`. The two-2s witness can also be checked independently:

```
python3 verify/verify2048.py witness/131072-two-twos.txt
```

Executable witnesses and experiments. All code is in the repository [5]. `node test/solve.js 3x3` reproduces Table 1 and Table 2 (3×3 takes about five minutes, $2 \times 2 \times 2$ fourteen seconds, and `FORWARD_ONLY=1 ... 2x5` three hours); `PERFECT=1 node test/run.js` replays each shipped line through the headless runner and asserts the exact counts (`GOAL=spiral|score`, `ORIENT=col`); `python3 verify/verify2048.py witness/131072.txt` checks a witness independently of everything else in the repository, and `node test/replay_engine.js witness/131072.txt` replays it through the original engine (`node test/gen_witness.js` regenerates the witnesses from the shipped data); `node test/paper_facts.js` checks Lemma 5, Proposition 15 and Theorem 22 against the shipped lines and, exhaustively, the tiny boards; `node test/gen_perfect.js` regenerates a line from nothing; `node test/honest.js` and `node`

`test/honest_drive.js` produce the honest-play table. The board drawings are generated from the engine’s own snake tables.

AI-assisted development

Generative AI assisted with conjecture exploration, manuscript drafting, programming, and Lean proof development. Kernel verification applies to the named formal statements under the displayed definitions; executable replay applies to the supplied finite witnesses. Neither constitutes peer review of the unformalized arguments or establishes historical priority. The proof and experiment sources are provided for independent checking.

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